



Course Materials

Chainsaw Maintenance & Cross Cutting

Course	Chainsaw Maintenance & Cross Cutting
Provider	Chainsaw Courses Ltd · chainsawcourses.com
Delivery	Fully Online — E-Learning (self-paced)
Hours	4 hrs GLH · 2 hrs Assessment · 4 hrs Self-Study · 10 hrs TQT
CPD	5 CPD Points (IIRSM)
Threshold	80% pass — randomised 45-question summative examination
Purpose	Training presentation & delegate materials

Contents

- 1 How to Access the Live App & Admin Panel
- 2 A Note on the Training Manual
- 3 Learning Outcome Framework — 3 Units · 6 LOs · 22 Assessment Criteria
- 4 Syllabus Mapping Table — 36 Modules · LO & AC Alignment
- 5 Compliance Tools & Practical Worksheets
- 6 Video Transcripts — 32 Modules
- 7 Assessment Bank — 151 Multiple-Choice Questions
- 8 Supplementary Oral & Practical Mock Questions (79)

SECTION 1

How to Access the Live App & Admin Panel

All content unlocked for IIRSM assessors

The course is delivered through a custom-built e-learning platform. All video modules, quizzes, AI-assisted mock practice, summative examination, and certificate generation are accessible via the two routes below.

Admin / Assessor Access — All Content Unlocked

- 1 Go to app.chainsawcourses.com
- 2 Scroll to the bottom of the screen and click the 'Admin Panel' button.
- 3 Enter the admin credentials when prompted — Password: CHAIN26 — the full admin dashboard will appear.
- 4 Explore the dashboard tabs: Feedback, Backup, Storage, Policy Documents and more.
- 5 All Export buttons send data directly to protected Google Drive storage.
- 6 Click 'APP PREVIEW' in the top-right corner of the dashboard to enter the fully unlocked app.
- 7 All videos, features, certificates, and IIRSM-referenced content are fully accessible for review.

New Learner Journey — Standard Access Route

Experience the app exactly as a learner — sequential locking, quiz flow, and liability waiver. The app is under internal testing and has not been publicly released.

- 1 Log out of the admin panel (or open an incognito browser window).
- 2 Go to app.chainsawcourses.com and enter login code: CHAIN
- 3 Enter a name and email address of your choice.
- 4 You will be directed to the Liability Waiver — accept to enter the course.
- 5 Each module is locked until the previous video is fully watched and the quiz completed.
- 6 On completion, the 45-question exam unlocks. 80% pass auto-generates the IIRSM certificate.

SECTION 2

A Note on the Training Manual

Published author · File size & access options

Published Author

Author of 'The Chainsaw Manual' (Version 1.1, July 2026) — currently sold as a standalone physical learning aid to various colleges and training providers across the UK. The manual underpins the theoretical content of this eLearning course. The first edition was published in September 2024.

The training manual — Chainsaw Maintenance & Cross Cutting: A Comprehensive Technical Manual (v1.1) — is the primary delegate learning resource. It is a richly illustrated 138-page document covering all Learning Outcomes and Assessment Criteria, including Advanced Workshop Extension modules.

File Size Notice

The manual PDF is approximately 71 MB due to high-resolution diagrams and photographs. This exceeds most standard upload limits and cannot be attached directly to this IIRSM submission.

How to obtain the manual:

Option A — Request a Digital Copy via WeTransfer (Recommended)

Email info@chainsawcourses.com to request a digital copy. A download link will be sent via WeTransfer or a similar large-file transfer service.

Option B — View within the App

The full manual is embedded within the course platform. After logging in via the Admin Access route in Section 1, click APP PREVIEW. The manual is viewable within the app at any point during the course. All supporting policy documents are also accessible via the Admin Dashboard.

Option C — Request a Physical Copy

A physical copy of the manual can be requested by contacting David Daniel at chainsawcourses.com.

SECTION 3

Learning Outcome Framework

3 Units · 6 Learning Outcomes · 22 Assessment Criteria

The course is structured across three units and six formal learning outcomes, each broken into specific assessment criteria (ACs). All examination questions in Section 7 are tagged to their LO and AC. The modules listed under each outcome are the primary digital activities through which the content is delivered.

LO1

Legal Framework & Personal Safety

Unit 1 — Occupational Standards, Health & Safety, and Risk Evaluation

Describe the statutory legal framework and personal safety requirements governing chainsaw operations.

- AC 1.1** Identify employer and employee obligations under the Health and Safety at Work Act (HSWA) and global equivalents.
- AC 1.2** Explain PUWER operational parameters governing tool maintenance and operator competence.
- AC 1.3** Summarise COSHH control tracking required for hazardous fuels, battery cells, lubricants, and toxic flora species.
- AC 1.4** Detail CE/UKCA and global standard class markings for safety helmets, hearing protection, gloves, and Type A/C protective trousers.

Activities: Equipment List · PPE & First Aid · Law & Regulations · Hazards & Risks

LO2

Hazard Evaluation & Emergency Protocols

Unit 1 — Occupational Standards, Health & Safety, and Risk Evaluation

Evaluate environmental hazards, risk metrics, and implement emergency protocols for chainsaw site operations.

- AC 2.1** Execute a 5-step site-specific risk assessment documenting ground hazards, pedestrian proximity, and structural vulnerabilities.
- AC 2.2** Formulate an emergency communication and extraction map with grid references, postcodes, access limitations, and trauma kit deployments.
- AC 2.3** Identify bio-security cleaning controls necessary to stop the spread of invasive arboreal pathogens and pests.

Activities: 5 Steps To Risk Assessment · Hazards & Risks · Emergency Planning Information

LO3

Chainsaw Architecture & Safety Features

Unit 2 — Power Unit Architecture, Mechanical Integrity, and Component Maintenance

Analyse the mechanical differences, design attributes, and safety features of internal combustion and battery-powered chainsaws.

- AC 3.1** Describe the 2-stroke combustion cycle and determine fuel-to-oil lubrication ratios at a standard 50:1 mix.
- AC 3.2** Compare the advantages and operational risks of battery-powered power units against internal combustion platforms.
- AC 3.3** Map and explain the mechanical function of the 10 core safety features across the front, centre, and rear chainsaw architecture.

Activities: Chainsaw Safety Features · Battery Chainsaws · Chain Brake · Fuel & Oil Filters

LO4**Diagnostic, Servicing & Maintenance Procedures**

Unit 2 — Power Unit Architecture, Mechanical Integrity, and Component Maintenance

Describe the diagnostic, servicing, and maintenance procedures required to sustain the structural integrity of the chainsaw cutting assembly.

AC 4.1 Detail air filter cleaning procedures and interpret spark plug electrode colour indicators (Brown / Black / White-Grey).

AC 4.2 Explain safe carburettor adjusting rules using factory Idle (LA/T), Low (L), and High (H) screw limit constraints. (Advanced Extension)

AC 4.3 Differentiate Rim and Spur drive sprockets and diagnose guidebar wear including burring, rail splaying, and thermal bluing.

AC 4.4 Identify chain pitch, gauge, and tooth shapes (Full-Chisel vs. Semi-Chisel) and calculate correct filing profile configurations.

Activities: Air Filter · Spark Plug · Cooling System · Exhaust · The Oiling System · Recoil Starter · Clutch Assembly · Sprocket · Guidebar · Chain Basics · Chain Tension · How to identify a chainsaw chain · Replacing The Chain · Chain Sharpening

LO5**Pre-Use Verification & Startup Methodologies**

Unit 3 — System Startups, Operational Testing, and Processing Techniques

Implement safe pre-use verification protocols and startup methodologies for safe chainsaw deployment.

AC 5.1 Differentiate between safe cold start floor-anchor and upright knee-clamp warm start methods for Husqvarna and Stihl platforms.

AC 5.2 Perform a 4-point dynamic check assessing chain brake engagement, oil dispersion flow, chain creep at tick-over, and off-switch motor cut.

Activities: Pre-Start Checks · Starting The Chainsaw · Pre-Use Checks

LO6**Cutting Mechanics: Tension & Compression**

Unit 3 — System Startups, Operational Testing, and Processing Techniques

Apply mechanical principles to resolve tension and compression forces during timber cross-cutting operations.

AC 6.1 Analyse a log setup to determine where tension and compression forces reside.

AC 6.2 Explain the physics behind pulling chains versus pushing chains during cross-cutting.

AC 6.3 Define the bar tip kickback zone and explain how to execute precise plunge bore entries safely.

AC 6.4 Differentiate cut sequences for standard logs, oversized timber, and timber under extreme tension (Toast Rack / reduction sink).

AC 6.5 Evaluate specialised branch removal, snedding, and de-limbing sequences along a felled stem. (Advanced Extension)

AC 6.6 Identify site threats with windblown windfalls, establishing escape routes and managing root plate movements. (Advanced Extension)

Activities: Kickback · Work Positioning · Cutting Basics · Tension & Compression · Releasing A Trapped Chainsaw · Bore Cutting · Oversized & Tensioned Timber · Stacking · Additional Cuts

SECTION 4

Syllabus Mapping Table

36 Modules · Learning Outcome & Assessment Criteria Alignment

The table below maps every active app module to its corresponding Learning Outcome (LO) and Assessment Criteria (AC). This mirrors the alignment table at the back of the training manual. Modules categorised as Course Requirements are reference documents and do not carry a formal LO/AC assignment.

#	Module Title	Type	Category	LO	Assessment Criteria	Manual pp.
COURSE REQUIREMENTS — REFERENCE DOCUMENTS						
1	Equipment List	App	Course Req.	—	—	—
UNIT 1 — OCCUPATIONAL STANDARDS, HEALTH & SAFETY & RISK EVALUATION						
2	PPE & First Aid	Video	Assessment	LO1	AC 1.4	pp.10–21
3	5 Steps To Risk Assessment	Video	Assessment	LO2	AC 2.1	pp.14–15
4	Hazards & Risks	App	Assessment	LO1, LO2	AC 1.3, AC 2.3	pp.17–23
5	Emergency Planning Information	Video	Assessment	LO2	AC 2.2	p.16
6	Law & Regulations	Video	Assessment	LO1	AC 1.1, AC 1.2	p.18
7	Chainsaw Safety Features	Video	Assessment	LO3	AC 3.3	pp.30–33
UNIT 2 — POWER UNIT ARCHITECTURE, MECHANICAL INTEGRITY & COMPONENT MAINTENANCE						
8	Battery Chainsaws	Video	Maintenance	LO3	AC 3.2	pp.34–35
9	Air Filter	Video	Maintenance	LO4	AC 4.1	pp.36–39
10	Spark Plug	Video	Maintenance	LO4	AC 4.1	pp.36–39
11	Cooling System	Video	Maintenance	LO4	AC 4.1	pp.36–39
12	Exhaust	Video	Maintenance	LO4	AC 4.1	pp.36–39
13	Fuel & Oil Filters	Video	Maintenance	LO4	AC 4.1, AC 3.1	pp.26–39
14	The Oiling System	Video	Maintenance	LO4	AC 4.1	pp.36–39
15	Recoil Starter	Video	Maintenance	LO4	AC 4.1	pp.36–39
16	Clutch Assembly	Video	Maintenance	LO4	AC 4.1	pp.36–39
17	Sprocket	Video	Maintenance	LO4	AC 4.3	pp.54–63
18	Chain Brake	Video	Maintenance	LO3	AC 3.3	pp.30–33
19	Guidebar	Video	Maintenance	LO4	AC 4.3	pp.54–63
20	Chain Basics	Video	Maintenance	LO4	AC 4.4	pp.64–81
21	Chain Tension	Video	Maintenance	LO4	AC 4.4	pp.64–81
22	How to identify a chainsaw chain	Video	Maintenance	LO4	AC 4.4	pp.64–81
23	Replacing The Chain	Video	Maintenance	LO4	AC 4.4	pp.64–81
24	Chain Sharpening	Video	Maintenance	LO4	AC 4.4	pp.64–81
UNIT 3 — SYSTEM STARTUPS, OPERATIONAL TESTING & PROCESSING TECHNIQUES						
25	Kickback	Video	Cross Cutting	LO6	AC 6.3	pp.82–83
26	Work Positioning	Video	Cross Cutting	LO6	AC 6.1	pp.104–107
27	Pre-Start Checks	Video	Cross Cutting	LO5	AC 5.2	pp.100–101
28	Starting The Chainsaw	Video	Cross Cutting	LO5	AC 5.1	pp.96–99
29	Pre-Use Checks	Video	Cross Cutting	LO5	AC 5.2	pp.100–101
30	Cutting Basics	Video	Cross Cutting	LO6	AC 6.1	pp.104–107
31	Tension & Compression	Video	Cross Cutting	LO6	AC 6.1, AC 6.2	pp.104–107
32	Releasing A Trapped Chainsaw	Video	Cross Cutting	LO6	AC 6.2	p.104

#	Module Title	Type	Category	LO	Assessment Criteria	Manual pp.
33	Bore Cutting	Video	Cross Cutting	LO6	AC 6.3	pp.110–111
34	Oversized & Tensioned Timber	Video	Cross Cutting	LO6	AC 6.4	pp.108–121
35	Stacking	Video	Cross Cutting	LO6	AC 6.4	pp.108–121
36	Additional Cuts	Video	Cross Cutting	LO6	AC 6.4	pp.108–121

Key

LO1–LO6	Learning Outcomes 1–6 across three units of the qualification
AC x.x	Assessment Criteria — each LO has 2–6 specific criteria
—	Course Requirements modules are prerequisite reference docs; no formal LO/AC assigned
Video	Interactive video module with embedded quiz (sequentially gated)
App	Content embedded in the app interface (e.g. main page, risk assessment page) — not a separate file
PDF	Read-only reference document (does not gate progression)

SECTION 5

Compliance Tools & Practical Worksheets

Available within the course app · PPE Verification · Risk Assessment · Emergency Action Plan · Bio-Security · CPD News

All practical compliance worksheets are embedded directly within the course app and are completed digitally by the learner. This approach ensures records are date-stamped, stored securely, and immediately exportable as a PDF or plain-text report for portfolio submission or regulatory inspection.

PPE Verification Checklist

Accessible from the app Inspection Checklist. The learner confirms each item of PPE — helmet, hearing protection, gloves, trousers, boots, hi-vis, and first aid kit — against its UK/international standard (EN 397, EN 352, EN ISO 11393, EN ISO 17249, EN ISO 20471, BS 8599-1). Mapped to LO1 / AC 1.4 and LO2 / AC 2.2.

Pre-Start & Pre-Use Chainsaw Inspection

Also within the app Inspection Checklist. Covers chain tension, sharpness, brake function, bar condition, oiler, fuel mix, air filter, handguards, chain catcher, exhaust, anti-vibration mounts, and controls. A 4-point dynamic check confirms brake engagement, oil spray, chain creep at idle, and off-switch cut. Mapped to LO5 / AC 5.2.

Risk Assessment, Emergency Action Plan & Bio-Security Log

A 5-step site risk assessment template, Emergency Action Plan (site grid reference, nearest A&E, first aider, muster point), and bio-security cleaning log are available via the app dashboard. Each record is retained and can be exported for IIRSM portfolio use. Mapped to LO2 / AC 2.1, 2.2, and 2.3.

News & CPD Updates

Industry news is published regularly through the app and tagged to its relevant Learning Outcome and AC. Engaging with updates directly supports the IIRSM RM&CF Ongoing Competence Assurance pillar. Accessible via the Admin Dashboard under the Updates tab.

SECTION 6

Video Transcripts

32 Modules · LO & AC Aligned · Timecoded

The transcripts below correspond to each video module in the course. They are provided for assessor and IIRSM review purposes only and are not accessible to learners within the platform. Each transcript is mapped to its Learning Outcome (LO) and Assessment Criteria (AC) and includes timecodes from the original recording.

Module 2: PPE & First Aid		LO1 · AC 1.4
Timecode	Transcript	
0:01 – 0:09	In this video, we'll cover the personal protective equipment, PPE for short, and some first aid provisions you'll need for operating a chainsaw.	
0:10 – 0:14	Let's start with essential personal protective equipment.	
0:14 – 0:19	You'll need chainsaw protective boots with steel or composite toe	
0:20 – 0:24	chainsaw trousers,	
0:24 – 0:28	gloves	
0:28 – 0:34	a suitable helmet with ear and I protection.	
0:34 – 0:42	In addition to the face shield, supplementary eye protection is recommended.	
0:42 – 0:46	Chainsaw PPE can be identified with this symbol	
0:46 – 0:49	and is available in four main categories.	
0:49 – 1:17	Class zero 16m A second. Class one 20m A, second class two 24m a second and class three 28m a second. Chainsaw protection is made up of layers of ballistic nylon, Kevlar and other synthetic materials. It is designed to clog up the sprocket to stop the chain moving, but do not assume this will always work due to the difference in chainsaw power.	
1:17 – 1:30	Carry out a risk assessment to decide what level of chainsaw protection you need. For example, using more powerful chainsaws will require a higher level of protection.	
1:30 – 1:33	You also need a personal first aid kit	
1:34 – 1:38	with a minimum of a large trauma, wound dressing,	
1:42 – 1:46	ensure you have received training in how to use your first aid kit.	
1:47 – 1:51	Have a larger, more comprehensive first aid kit nearby	
1:51 – 1:56	a nominated first aider.	
1:56 – 2:00	This concludes the video on PPE and first aid requirements.	

Module 5: Emergency Planning Information		LO2 · AC 2.2
Timecode	Transcript	
0:01 – 0:04	This video is about emergency planning and information.	
0:05 – 0:24	Emergency planning is vital for everyone's safety on site. In an event of a major incident occurring, you need to be prepared for the worst. The following list is not exhaustive, but highlights the more important information that needs to be recorded. The location with street address and postcode. A grid reference or what?	
0:24 – 0:28	Three words. The nearest hospital with its phone number.	
0:28 – 0:31	Site access and extraction	
0:31 – 0:35	a rendezvous point to meet the emergency services.	
0:35 – 0:41	The nearest phone with signal. Emergency contact details of all personnel on	
0:41 – 0:53	The site manager's contact. Potential helicopter landing area. The location of the team. First aid kit. The nominated first aider and any useful preexisting medical conditions.	
0:53 – 1:03	Have this information available to all on site and keep it in a dry, easy to access location.	
1:03 – 1:06	This concludes the video on emergency planning.	

Module 6: Law & Regulations		LO1 · AC 1.1
Timecode	Transcript	
0:01 – 0:09	In this video, we will cover health and safety law and basic regulations in relation to chainsaw maintenance and crosscutting.	
0:09 – 0:21	In 1974, the Health and Safety at Work Act became enforceable by law. This required that the employer and the employees have a duty of care to maintain a safe working environment,	
0:21 – 0:26	follow any training given, and wear any appropriate PPE.	
0:26 – 0:36	Regulations were put in place to help employers adhere to the law, of which there are many. The main regulation for chainsaw operation is PUWER	
0:36 – 1:03	the provision and use of work equipment regulations. This requires that all equipment, including chainsaws, are well maintained, are fit for purpose and operators must be trained and competent. Organizations such as the AA, the Arboriculture Association and FISA, the Forestry Industry Safety Accord and regional forestry bodies give industry guidance for tree work.	
1:03 – 1:33	We also need to adhere to various regulations covering the protection of wildlife and their habitats in order to minimise environmental damage. We can choose fuelling areas away from watercourses, use spill mats, consider using battery chainsaws and use environmentally friendly of fuels. And you also need to think about biosecurity measures like disinfecting PPE and equipment to ensure no cross-contamination of pests and diseases.	
1:33 – 1:36	This concludes the video on the law and regulations.	

Module 7: Chainsaw Safety Features		LO3 · AC 3.3
Timecode	Transcript	
0:01 – 0:17	In this video, we'll look at the ten main safety features on a chainsaw. These have been designed to minimise risk and protect the user. We'll make sure that all safety features are present, intact and working properly with a brief description on each.	
0:21 – 0:27	To make things easier, we'll look at the safety features in different sections.	
0:27 – 0:31	Firstly, we have four at the	
0:31 – 0:35	then two in the middle	
0:35 – 0:39	and four at the back.	
0:39 – 0:51	If you look at the saw at the front, you can see four safety features straight away.	
0:51 – 0:55	The first is the chain brake and left hand guard.	
0:55 – 1:05	This stops the chain rotating and helps protect the left hand when kickback occurs. The chain brake is designed to be activated by the back of the left hand.	
1:05 – 1:21	It can also be activated from inertia and abrupt change in motion. Similar to how your seatbelt works. Refer to the section on kickback for more information. As kickback is a main contributing factor for chainsaw accidents.	
1:21 – 1:26	Just underneath the chain brake, we have the exhaust.	
1:26 – 1:32	This directs the fumes away from the operator and reduces the noise and emissions.	
1:32 – 1:37	Next we have the bar and chain combination.	
1:37 – 1:46	These need to be compatible with each other and the chain should have kickback reducing features.	
1:46 – 1:50	The fourth and final safety feature at the front	
1:50 – 2:00	is the scabbard. This protects the user when handling the chainsaw and protects other equipment from damage when in transportation.	
2:00 – 2:02	the middle of the chainsaw,	
2:02 – 2:07	have another two safety features.	
2:07 – 2:12	we turn the saw upside down, we can now clearly see the two safety features present.	
2:12 – 2:19	They are one of the anti-vibration mounts and the chain catcher.	
2:19 – 2:24	These anti-vibration mounts are located throughout the saw,	
2:24 – 2:30	normally a mixture of springs and rubbers to help minimise the effects of vibration.	

2:30 – 2:41	Hand arm vibration syndrome or HAVS can be serious and it can lead to musculoskeletal injuries and other nervous system issues.
2:41 – 2:45	The next one is the chain catcher.
2:45 – 2:49	This catches the chain when either the chain snaps or derails.
2:49 – 3:04	The chain derailing happens more than you think. The chain catcher is made of a softer metal or plastic to grip the chain as it comes off the guide bar. As it does so, the momentum of the chain whips round to the rear of the saw.
3:04 – 3:10	This is where we'll find our next safety feature. The rear or right hand
3:10 – 3:21	This protects the right hand when the chain snaps or derails. Normally on top or near to the right hand guard are the safety stickers or decals.
3:21 – 3:32	These provide safety information and warnings. Icons highlighted in blue are mandatory actions you need to take, and those in yellow are warnings.
3:32 – 3:36	We then have the throttle
3:36 – 3:42	This two trigger mechanism stops any unwanted depression of the trigger.
3:42 – 3:45	finally we have the on off switch.
3:45 – 3:52	This needs to be operational, allowing the user to stop the engine at any given moment.
3:52 – 4:04	If one of these safety features is missing or faulty, the chainsaw should be taken out of action and repaired immediately. The saw needs to be labelled as not safe to use until it's repaired.
4:04 – 4:08	This concludes the video on the ten safety features of the chainsaw.

Module 8: Battery Chainsaws

LO3 · AC 3.2

Timecode	Transcript
0:01 – 0:05	This video covers battery chainsaws, and the charging elements.
0:05 – 0:10	Firstly, let's look at the advantages of using battery chainsaws.
0:10 – 0:25	They can be lighter, quieter, reduce vibrations, easier to control and with no emissions to the user. There's no fuel to transport and to spill when refueling and communication on site is easier.
0:25 – 0:29	Now let's look at some hazards. With the use of battery chainsaws.
0:30 – 0:47	They can be heavier and lack sufficient power. They can malfunction and short due to water ingress and carry the risk of electric shock. The batteries need to be charged, stored and disposed of correctly and can potentially combust when charging.
0:47 – 0:50	How do you maintain the battery elements?
0:51 – 0:56	Initially, start by unplugging and removing any batteries or charges.
0:56 – 1:08	Inspect batteries and charges for cracks and are not damaged in any way.
1:08 – 1:16	Make sure all battery tracks and contact points are dry and clean. Use a brush or airline to clean if dirty.
1:16 – 1:24	If in any doubt, retire as they can become a potential fire hazard.
1:24 – 1:28	This concludes the video on battery chainsaws and charging elements.

Module 9: Air Filter

LO4 · AC 4.1

Timecode	Transcript
0:01 – 0:05	In this video, we'll explore the air filter
0:05 – 0:08	First of all, where is the air
0:08 – 0:11	The air filter is found under the top cover
0:11 – 0:17	towards the rear of the machine.
0:17 – 0:26	The top cover is usually attached by screws, bolts, or catches and there are usually three of
0:26 – 0:29	What does it

0:29 – 0:42	The air filter keeps dust and debris out of the carburetor and helps to regulate the air fuel mixture. Air filters come in various shapes, sizes, and materials, but all do the same job.
0:42 – 0:55	How do you maintain it? Air filter removal is straightforward. Some are attached with clips or screws. Others remove easily with just a twist or pull.
0:55 – 0:59	Clean the area around the air filter and carburetor.
0:59 – 1:04	Close the butterfly valve by applying the choke setting.
1:04 – 1:15	Remove the main air filter and pre filter if present, and fill the void with a clean rag or similar. This is to stop any debris falling into the carburetor when cleaning.
1:15 – 1:21	Check the air filter for any rips, tears, holes, or any discoloration.
1:21 – 1:26	Tap out any loose dust and use a soft brush or pressurized airline to clean.
1:26 – 1:50	Pressurized air must be directed from the inside out and not too close, as this may damage the filter. Some air filters separate into two parts to aid cleaning. Some air filters can be washed with warm soapy water. Leave to completely dry before refitting. If in any doubt of the condition of the air filter, replace it as it is the lungs of your machine.
1:50 – 1:56	Screw on or twist to reinstall.
1:56 – 1:58	This concludes the video on the air filter.

Module 10: Spark Plug		LO4 · AC 4.1
Timecode	Transcript	
0:02 – 0:06	This video is all about the spark	
0:06 – 0:16	Where is the spark plug? The spark plug is located on top of the cylinder.	
0:16 – 0:18	What does it do?	
0:19 – 0:24	It produces a spark to ignite the air fuel mixture in the cylinder.	
0:24 – 0:26	How do you maintain	
0:26 – 0:29	Firstly pull the lead off the spark plug.	
0:29 – 0:35	and unscrew the spark plug using the correct sized combi spanner or socket.	
0:35 – 0:42	On some models, you may have to push through a protective rubber to get to the spark plug.	
0:42 – 0:45	Now we can check a few things.	
0:45 – 0:53	Some spark plugs have a screw top where the lead fits. Check for tightness.	
0:53 – 1:02	Examine the white ceramic for any discoloration, cracking, or damage which may cause the spark to arc.	
1:02 – 1:09	Check the compression washer for cracks or where	
1:09 – 1:14	the electrode gap is usually set at 0.5mm.	
1:14 – 1:21	Check with a feeler gauge and adjust if needed.	
1:21 – 1:34	If there is an accumulation of carbon or corrosion on the electrode, gently clean with a brass wire brush. If the spark plug displays any damage, then simply replace it as they are inexpensive.	
1:34 – 1:36	Check the electrode	
1:36 – 1:45	This should be a light brown color. This indicates the fuel air mix is correct.	
1:45 – 1:50	If it is too black or dark, it could indicate a blocked air filter	
1:50 – 1:55	or too much two stroke oil in the fuel.	
1:55 – 2:01	If it is too gray or white, the chainsaw is running hot and could indicate a dirty fuel filter.	
2:01 – 2:04	A hole in the air filter	
2:04 – 2:08	not enough to stroke in the fuel	
2:08 – 2:10	to reinstall the spark plug.	
2:10 – 2:22	Screw in by hand. Being careful not to cross thread and gently tighten with a socket. Do not over tighten as this can damage the thread washer or cylinder block.	

2:22 – 2:27	Remember to reinstall the lead.
2:27 – 2:29	This concludes the video on the spark plug.

Module 11: Cooling System		LO4 · AC 4.1
Timecode	Transcript	
0:01 – 0:06	This video is about the cooling system, which uses airflow for cooling.	
0:06 – 0:15	Where is the cooling system? The cooling system is comprised of the flywheel and the cylinder fins.	
0:15 – 0:23	The flywheel is found underneath the recoil side cover, and the cylinder fins can found under the main top cover.	
0:23 – 0:30	What does it do? Its function is to cool the main cylinder block and prevent overheating of the engine.	
0:30 – 0:45	As the flywheel spins, it draws air in up, across and through the cylinder fins, allowing a constant flow of air. Some air is also fed to the air filter.	
0:45 – 0:48	How do you maintain it?	
0:48 – 0:55	Unscrew the four bolts on the recoil housing on some saws. You will have to remove the top cover to access one of the bolts.	
0:55 – 0:59	Check for damage and clean all vents.	
0:59 – 1:09	Inspect the flywheel and cylinder fins for any cracks, dense missing fins or build up of debris and clean them with a soft brush or airline.	
1:09 – 1:16	Avoid using any metal tools on the fins as this could damage them.	
1:16 – 1:23	If flywheel or cylinder fins are damaged or missing, replace immediately	
1:24 – 1:31	before putting the cover back on. See the video on the recoil assembly.	
1:31 – 1:34	This concludes the video on the cooling system.	

Module 12: Exhaust		LO4 · AC 4.1
Timecode	Transcript	
0:00 – 0:05	In this video, we'll look at the exhaust.	
0:05 – 0:07	Where is the exhaust?	
0:07 – 0:11	The exhaust is generally located at the front of the machine,	
0:11 – 0:13	below the chain brake.	
0:13 – 0:16	What does it do?	
0:16 – 0:26	The exhaust reduces the noise and emissions of the engine. It is also designed so that the fumes are pushed away from the operator.	
0:26 – 0:29	How do you maintain	
0:29 – 0:29	Check.	
0:29 – 0:42	The bolts are tight and secure. Examine for any dense cracks or holes. Clean any debris from in and around the exhaust area with a brush or airline.	
0:42 – 0:46	Some chainsaws have an external spark arrestor.	
0:46 – 0:53	Periodically remove and clean with a wire brush.	
0:53 – 1:01	If the exhaust is damaged, replaced with a new one and always change the gasket.	
1:01 – 1:04	This concludes the video on the exhaust.	

Module 13: Fuel & Oil Filters		LO4 · AC 4.1
Timecode	Transcript	
0:01 – 0:05	In this video, we'll cover the fuel and all filters.	
0:05 – 0:08	We'll begin with the fuel	

0:08 – 0:11	So where is the fuel filter?
0:11 – 0:16	The fuel filter is found inside the fuel tank attached to a black hose.
0:16 – 0:21	There are different designs, but they all have the same function.
0:21 – 0:23	What does it do?
0:23 – 0:31	The fuel filter filters out debris in the fuel from entering the carburetor and cylinder, ensuring a clean fuel supply.
0:32 – 0:41	How do you maintain it? Firstly, empty any fuel in the tank into an approved container.
0:41 – 0:46	Use a wire hook to fish out the filter. Don't use pliers or mole grips
0:46 – 0:49	as these could damage the rubber hose.
0:49 – 1:01	Inspect for any rips, holes or damage to the filter or hose. If any defects or heavy discoloration are found, replace the filter as they are inexpensive.
1:01 – 1:18	When replacing the filter, make sure you have the new filter in arm's reach. Then just pull the old one off and pop the new one
1:18 – 1:22	Now let's move on to the filter.
1:22 – 1:27	The oil filter is found inside the oil tank.
1:27 – 1:37	The majority of oil filters are either fixed into the main body or located on a short hose, so cannot be removed or cleaned easily.
1:37 – 1:39	What does it do?
1:39 – 1:47	The oil filter reduces debris from entering the oil system, ensuring clean oil is fed into the bar.
1:48 – 1:50	How do you maintain it?
1:50 – 1:54	Empty any point in the tank into an approved container.
1:54 – 1:58	Pour in the manufacturer's recommended degreaser.
1:58 – 2:11	Swirl it around. Leave it for a few minutes and swirl again. Then empty the degreaser. Repeat as necessary.
2:11 – 2:18	Alternatively, the filter can be accessed by removing the clutch assembly to expose the oil pump and system.
2:18 – 2:25	Refer to the additional content section and the video on the system for more help.
2:26 – 2:32	Find the rubber hose that leads into the reservoir and gently pull through the body of the chainsaw.
2:32 – 2:39	Inspect and replace both the filter and the hose if necessary.
2:39 – 2:42	This concludes the video on the fuel and filters.

Module 15: Recoil Starter		LO4 · AC 4.1
Timecode	Transcript	
0:00 – 0:06	This video is about the recoil assembly and how to loosen and tension the pull cord.	
0:06 – 0:13	Well, where is it? It is found on the opposite side of the chainsaw to the chain and bar.	
0:13 – 0:15	What does it do?	
0:15 – 0:19	When the handle is pulled. The flywheel is engaged by small pulls and springs,	
0:19 – 0:22	Either on the flywheel.	
0:22 – 0:24	or the recoil housing,	
0:24 – 0:28	allowing for the starting of the engine.	
0:28 – 0:30	How do you maintain it?	
0:30 – 0:38	Undo the four bolts on the cover. On some saws, you will have to remove the top cover to access a hidden bolt.	
0:39 – 0:47	Once off, inspect the cover and mechanism for any damage or buildup of debris and clean the vents with an airline or brush.	
0:47 – 0:51	How to detension the spring and pull cord.	
0:51 – 0:55	Firstly remove the plastic insert if there is one.	

0:55 – 1:01	Pull out the cord as far as you can and with the other hand, hold the mechanism with your thumb.
1:01 – 1:06	Be careful at this point, since the cord can whip back towards your hand if you lose your grip.
1:06 – 1:20	Pinch the cord just inside the cover. Pull through the entire length and hold directly vertical. Now release your thumb grip. This should allow the spring to detention and the cord should be slack
1:20 – 1:27	on some source. There will be a notch that the cord has to be placed into.
1:27 – 1:35	Pull the cord back through the hole and inspect along the length of it for any damage.
1:35 – 1:41	In particular, check the knots at both ends
1:41 – 1:46	and where the cord rubs against the handle and side cover.
1:46 – 1:58	How do you retention the pull cord? Take a firm grip of the cord and start to wrap it clockwise around the mechanism. You need to turn the mechanism while doing this to tension the spring properly.
1:58 – 2:00	On some saws, you will need to locate the notch
2:00 – 2:23	before you start winding. Depending on how long your cord is, you will need to complete 4 to 6 turns. Hold the mechanism with your thumb and hold the handle with your other hand. Gently allow the handle to be drawn in by the spring. To check you have the correct tension, bend the handle down to the side and allow it to spring back up into position.
2:23 – 2:35	If this doesn't happen, you will need to add another wrap or two
2:35 – 2:46	to check you haven't over tightened it. Pull the cord out as far as it will go. Hold the mechanism with your thumb and twist the mechanism clockwise.
2:46 – 2:53	If you find this hard, you can pull the cord at full length and wrap around the handle.
2:53 – 2:56	It should easily move at least a quarter to half a turn more.
2:56 – 3:09	If it doesn't, you have over tightened it and will need to release a wrap or two. Once the correct tension has been achieved, simply place back onto the sole and tighten the bolts back into place.
3:09 – 3:12	This concludes the video on the recall assembly.

Module 16: Clutch Assembly		LO4 · AC 4.1
Timecode	Transcript	
0:01 – 0:05	This video is all about the clutch assembly.	
0:06 – 0:08	So where is the clutch assembly?	
0:08 – 0:13	The clutch assembly is located behind the chain and bar side cover.	
0:13 – 0:19	The clutch assembly has the same components, whether it's in an inboard	
0:19 – 0:22	or outboard configuration.	
0:22 – 0:30	It consists of the sprocket,	
0:30 – 0:37	the clutch drum,	
0:37 – 0:41	the needle bearing,	
0:41 – 0:50	and the clutch weights and springs.	
0:50 – 0:53	What does the clutch assembly do?	
0:53 – 1:11	The clutch assembly engages the clutch drum and sprocket. As the clutch assembly spins. The clutch weights overcome the springs and are thrown outwards by centrifugal force. Gripping the clutch drum, turning the sprocket and therefore the chain.	
1:11 – 1:15	How do you remove and maintain an inboard clutch assembly?	
1:15 – 1:22	This is fairly straightforward as you only need to remove the clip and washer.	
1:22 – 1:28	This allows you to remove the sprocket,	
1:28 – 1:30	clutch, drum	
1:31 – 1:33	and needle bearing.	
1:33 – 1:36	Check for any damage to the springs and weights.	
1:36 – 1:40	Over time, the springs can eat and wear into the weights.	

1:40 – 1:46	Check the clutch drum for any cracks blowing or any damage.
1:46 – 1:59	Check the needle bearing and apply grease as necessary. On some saws, this can be done without having to remove the clutch assembly using a small grease gun pumped into the middle.
1:59 – 2:07	If you need to remove the clutch weights and springs, you will need to remove the sparkplug and insert a piston stop.
2:07 – 2:21	This will allow you to undo it with a socket or spanner. This has an opposite thread, which means you have to turn it clockwise to loosen it.
2:21 – 2:29	When reinstalling the clutch drum and sprocket, make sure the oil notch lines up with the oil pump arm
2:29 – 2:31	and make sure it is properly seated.
2:31 – 2:39	If not, this can damage the oiling system.
2:39 – 2:42	Put the sprocket back on.
2:43 – 2:46	Pop the washer on
2:46 – 2:51	and use some pliers to snap the clip back into place.
2:51 – 2:56	To remove an outboard clutch, assembly is similar.
2:56 – 3:01	Remove the spark plug. Insert a piston stop
3:01 – 3:10	and using a special tool, undo the clutch assembly, turning clockwise to loosen.
3:10 – 3:21	Alternatively, you can place a combo spanner in one of the grooves and hit sharply with a hammer to shock the clutch off.
3:22 – 3:36	Once loosened, remove all the components. The clutch weights, clutch, drum, sprocket and needle bearing and perform the same maintenance checks as mentioned before.
3:36 – 3:38	This concludes the video on the clutch assembly.

Module 17: Sprocket		LO4 · AC 4.3
Timecode	Transcript	
0:01 – 0:04	In this video, we'll look at the sprocket.	
0:04 – 0:08	First of all, where is the sprocket?	
0:09 – 0:14	The sprocket is situated on or is part of the clutch drum?	
0:14 – 0:19	What does it do? The sprocket drives the chain around the bar.	
0:20 – 0:29	The drive links of the chain sit in the sprocket grooves, giving the chain its drive when the clutch is activated.	
0:29 – 0:33	There are two types of sprocket	
0:33 – 0:37	a rim sprocket	
0:37 – 0:40	a spur sprocket.	
0:40 – 0:48	A rim sprocket floats on the clutch drum and is cheaper to replace but wears quicker.	
0:48 – 0:56	A spur sprocket is integral with the clutch drum and lasts longer but costs a little more.	
0:56 – 1:03	How do you maintain it? Inspect the sprocket for any damage, cracks or excess where	
1:03 – 1:06	some sprockets have where indication lines.	
1:06 – 1:10	If you can't see a continuous line, the sprocket is to warn.	
1:10 – 1:18	If it doesn't, check with a gauge. The where should be no more than 0.5mm.	
1:18 – 1:30	If this is the case or it is damaged, replace immediately as a worn sprocket damages the chain and the guide bar. Watch the next video on how to remove the sprocket from the clutch assembly.	
1:30 – 1:32	This concludes the video on the sprocket.	

Module 18: Chain Brake		LO3 · AC 3.3
Timecode	Transcript	

0:00 – 0:04	In this video, we'll be looking at the chain break.
0:04 – 0:06	Where is the chain break?
0:06 – 0:10	The chain break is located at the front of the saw,
0:11 – 0:15	and consists of the left hand guard,
0:15 – 0:19	connected via a spring to the chain break band.
0:19 – 0:25	The chain break band is either housed in the side cover
0:25 – 0:28	or main body of the saw.
0:28 – 0:33	Depending on if you have an inboard or outboard clutch,
0:33 – 0:35	what does it do?
0:35 – 0:40	When applied, the chain break band tightens around the clutch drum,
0:40 – 0:51	stopping it from turning the sprocket, and therefore the chain chain breaks are activated manually with the back of the left hand, or are triggered by inertia.
0:51 – 1:06	How do you maintain it? When working on an outboard chain break band, the chain break must be deactivated to remove the side cover and once off, reactivated to help release the tension in the spring,
1:06 – 1:13	remove a few small screws and plastic cover to access the entire band.
1:13 – 1:18	Be careful when doing this as the spring can jump out.
1:18 – 1:29	Inspect the metal band for any bluing or overheating, distortion, cracks, and any damage. In particular, the thinnest points.
1:29 – 1:34	Clean any debris with an airline or brush and replace if damaged.
1:34 – 1:40	Add grease to keep the mechanism working smoothly.
1:40 – 1:46	Reinstallation of the spring can be tricky. It's mostly brute force when the spring is back in place.
1:46 – 2:01	The break will be in the activated position. Put the plastic cover on and screws back in, and deactivate the chain break by using the left hand guard. This will allow you to get the slide cover back on
2:01 – 2:10	to remove and inspect. A chain break band positioned around an inboard clutch is much the same. Make sure the chain brake is released.
2:10 – 2:22	Remove any screws or covers to reveal the spring and chain break mechanism.
2:22 – 2:34	Remove the band and inspect as before and clean if necessary before installing. Keep the chain brake well maintained as this will potentially save your life.
2:34 – 2:36	This concludes the video on the chain brake.

Module 19: Guidebar		LO4 · AC 4.3
Timecode	Transcript	
0:01 – 0:05	In this video, we'll be covering the guide bar.	
0:05 – 0:14	First of all, it might seem obvious, but where is it? The guide bar is located at the front of the chainsaw.	
0:14 – 0:16	What does it	
0:16 – 0:28	The guide bar provides the rails on which the chain sits, guiding the chain around the bar. It also allows the chain rule to circulate effectively and helps keep optimum chain cutting efficiency.	
0:29 – 0:40	How do you maintain it? First thing to check for is the straightness. Is the bar bent? If so, discard and replace immediately.	
0:40 – 0:45	Check the rail heights are even with a set square.	
0:45 – 0:49	Then check that the rails are not pinched,	
0:49 – 0:53	splayed,	
0:53 – 0:58	or overly	
0:58 – 1:05	You can file the rails to an even height using a flat file or a bar dresser.	
1:05 – 1:10	Inspect for burning, cracking, and any bluing of the metal caused by overheating	

1:10 – 1:19	check the nose sprocket, which should turn easily. Burning is where the metal wears down and folds over the side of the bar.
1:19 – 1:32	This can be filed off with a flat file filing away from the sprocket. If not removed, the burrs can lead to larger chunks being broken off the rails
1:32 – 1:38	and finally scrape out any debris sitting in the rails dragging away from the nose sprocket.
1:38 – 1:52	Clear the oil holes and check the rail heights. They should be a minimum of 4 to 6mm in depth when reinstalling. Flip the bar occasionally for a more even wear
1:52 – 2:08	the problems you might encounter. If all or some of these issues are present are inaccurate. Cutting, increased vibration, reduced oiling, overheating, and increased wear on the chain, guide bar, and sprocket.
2:08 – 2:11	This concludes the video on the guide bar.

Module 20: Chain Basics		LO4 · AC 4.4
Timecode	Transcript	
0:01 – 0:05	In this video, we'll look at the chainsaw chain basics.	
0:05 – 0:11	A chainsaw chain is made up of different components, each having their own function.	
0:11 – 0:15	Simply, we have the drive links,	
0:15 – 0:17	the cutter	
0:17 – 0:25	and the tie straps and rivets.	
0:25 – 0:30	The drive links sit in the sprocket	
0:31 – 0:41	and as it spins, they are driven around the guide bar, picking up chain on their way and transporting the oil around the bar.	
0:41 – 0:47	The cutters sever the timber	
0:47 – 0:48	the tie straps and rivets.	
0:48 – 0:54	Hold everything together and sit on the guide bar rails.	
0:54 – 0:57	The cutter does the hard work of severing the timber.	
0:57 – 1:02	So let's have an in-depth look.	
1:02 – 1:06	The first part to contact the timber is the depth gauge.	
1:06 – 1:14	This is sloped to reduce kickback and sets the depth of cut. Regulating the size of woodchip.	
1:14 – 1:20	too high and your cutter won't cut	
1:20 – 1:21	too low.	
1:21 – 1:27	this makes the chain very aggressive and dramatically increases kickback.	
1:28 – 1:30	Following right behind the depth gauge	
1:30 – 1:36	is the working corner.	
1:36 – 1:43	The working corner is first to start severing the timber and therefore needs to be sharp.	
1:43 – 1:48	the top plate and side	
1:48 – 1:57	Then work together to carve out a chip of	
1:57 – 2:04	which is then carried away in the gullet	
2:04 – 2:15	where the side and top plate meet. Determines how the cutter performs and is called the cutter profile.	
2:15 – 2:18	The two main profiles are full chisel	
2:18 – 2:23	and semi chisel.	
2:23 – 2:30	The full chisel has a 90 degree angle between the side plate and top plate.	
2:30 – 2:41	It is excellent at cutting softwood, but dulls quicker in harder timber and increases the risk of kickback.	
2:41 – 2:47	The semi chisel has a rounded angle where the side and top plates meet.	
2:47 – 2:52	It is used for hardwood and is slower cutting, but holds its edge longer	

2:53 – 2:56	It is a good choice for an all round chain, especially for those starting
2:56 – 3:01	since it reduces the risk of kickback.
3:01 – 3:03	This concludes the video on Chain Basics.

Module 21: Chain Tension		LO4 · AC 4.4
Timecode	Transcript	
0:00 – 0:05	This video covers the removal, refitting and tensioning of the chain	
0:06 – 0:13	Achieving the correct chain tension is important. If the chain is too loose, there is a risk of chain derailments.	
0:13 – 0:17	If too tight, it can overheat the chain and bar.	
0:17 – 0:23	Both of these scenarios will increase wear on the guide bar, sprocket and chain.	
0:23 – 0:26	So how do you tension the chain?	
0:27 – 0:38	Firstly, release the chain brake and use a combo spanner to turn the bolts anti-clockwise a few turns until loose. Do not try to tension the chain without loosening the side cover bolts first.	
0:38 – 0:43	This results in damage to the chain tensioner mechanism.	
0:43 – 0:49	This is found either in the sore body or the side cover.	
0:49 – 0:56	Once loosened, turn the tension a screw clockwise or anti-clockwise to achieve desired tension.	
0:56 – 1:04	The correct tension should allow the chain to rotate easily, but stop when force is released.	
1:04 – 1:11	Rest the nose of the bar on the work surface, then tighten the bolts back up	
1:11 – 1:15	Now let's go over the process of removing and refitting the chain and guide bar.	
1:15 – 1:28	When removing the chain and bar. Remember to always release the chain brake first if you don't, and try and remove the side cover, you can pull and distort the chain brake band.	
1:28 – 1:33	This only applies to chain saws with an outboard clutch assembly.	
1:33 – 1:38	Once the chain brake has been released, undo the bolts and turn the chain tensioner screw anti-clockwise.	
1:38 – 1:46	A few turns, then loosen the bolts all the way until you can remove the side cover.	
1:46 – 1:59	Bring the guide bar towards you and push towards the drive sprocket. This will allow the chain to slacken completely. Remove the chain from the nose before removing from behind the drive sprocket.	
1:59 – 2:07	Repeat the same method for an inboard clutch, but remove the chain from the drive sprocket and first	
2:07 – 2:10	to reassemble the chain and guide bar with an outboard clutch.	
2:11 – 2:19	Make sure you put the chain behind the clutch first and it is seated in the drive sprocket.	
2:19 – 2:23	Then manipulate the chain into the nose sprocket.	
2:23 – 2:28	It can be easier to have the chainsaw upright when doing this.	
2:28 – 2:36	if on its side, the chain can fall out the drive sprocket and groove, making things unnecessarily annoying.	
2:37 – 2:44	Put the side cover back on, making sure to line up the tensioner. Screw with the hole in the guide bar.	
2:44 – 2:51	If it's not lining up, undo the tensioner. Screw a little and move the bar back and forth until you feel it move into place.	
2:51 – 2:58	Now tighten the bolts up finger tight and turn the tensioner. Screw until the correct tension is achieved.	
2:58 – 3:04	Once this is done, firmly tighten up the bolts.	
3:04 – 3:25	If you have an even clutch, the only difference is to install the chain around the new sprocket first, and then manipulate the chain into the drive sprocket and make sure the guide bar is located into the chain tensioner.	
3:25 – 3:29	This concludes the video on chain tension removal and refitting.	

Module 22: How to Identify a Chainsaw Chain		LO4 · AC 4.4
Timecode	Transcript	
0:01 – 0:06	In this video, you'll learn how to identify a chainsaw chain.	

0:06 – 0:16	There are many manufacturers of chain and each have their unique markings or identification numbers. We will concentrate on the three main brands Husqvarna
0:16 – 0:17	Steel
0:17 – 0:21	and Oregon.
0:21 – 0:31	Firstly, determine what brand of chain you have. This can be found either on the tie strap or cutter link.
0:31 – 0:33	Here we have an Oregon chain
0:34 – 0:36	and this one a Husqvarna.
0:36 – 0:54	Both of these chains are identified in the same way. If you pull the chain up or out of the guide bar, you should see a number. This number then correlates to the specific chain on the chain chart. Identify what cutter profile you have a full or semi chisel,
0:54 – 0:59	And then in this example, we'll look for a Husqvarna 93
1:00 – 1:06	its corresponding file size and top plate filing angle.
1:07 – 1:18	will also tell you what pitch and gauge the chain
1:18 – 1:22	The chain will likely have another number below the depth
1:22 – 1:32	This number can either be the depth gauge setting or the pitch number. Don't make the mistake of using this number on the chain chart.
1:32 – 1:35	Identifying a still chain is slightly different.
1:35 – 1:38	Firstly, identify the number near the depth gauge,
1:39 – 1:51	In this case six. This number refers to the chain pitch. Then lift the chain up to find the number on the drive link, in this case three. This number refers to the chain gauge.
1:51 – 1:56	Together they make the number 63.
1:56 – 2:01	Now cross-reference this number to the chain
2:01 – 2:15	It will then tell you the file size, filing angle, and size of chain.
2:15 – 2:35	Sometimes the chain pitch is stamped under the depth gauge. Identify the number on the drive link as well, and find the corresponding pitch and gauge on the chart. This will give you the filing information you need.
2:35 – 2:39	This concludes the video on how to identify a chainsaw chain.

Module 23: Replacing The Chain		LO4 · AC 4.4
Timecode	Transcript	
0:01 – 0:05	In this video, we'll learn the information needed to replace the chain	
0:05 – 0:12	In order to replace the chain, we'll need five pieces of information, some of which can be found on the guide bar.	
0:12 – 0:20	the pitch, the gauge, the number of drive links, the length of guide bar, and the cutter profile.	
0:21 – 0:29	The pitch of the chain is its physical size. This is calculated by measuring the distance between any three rivets and divide by two.	
0:29 – 0:36	This measurement is important, otherwise you'll chain won't fit into your nose or drive sprocket.	
0:37 – 0:48	The gauge of the chain is the thickness of the drive link. This is also an important measurement since if it's too big, it won't fit inside the rails of the guide bar,	
0:48 – 0:54	if too small, we'll have excessive movement within the rails.	
0:54 – 1:02	The number of drive links and length of bar are essential as guide bars have slightly different shapes.	
1:02 – 1:11	And finally we have the cutter profile. This depends on the type of timber you're cutting or just your personal preference.	
1:11 – 1:14	This concludes the video on replacing the chain.	

Module 24: Chain Sharpening		LO4 · AC 4.4
Timecode	Transcript	
0:00 – 0:06	In this video, we'll learn how to sharpen a chainsaw chain.	20

0:06 – 0:13	Before we begin sharpening. We need gloves and I protection.
0:13 – 0:23	Now we need to identify what chain we have. If you're unsure how to do this, revisit the video on how to identify your chain.
0:23 – 0:28	The chain shown here is a still 23.
0:28 – 0:35	which means we need a 3/16 or 4.8mm file,
0:35 – 0:42	we're filing at a 30 degree angle.
0:42 – 0:52	We then need to select the appropriate file guide. There are many types of file guide which work in different ways, but all try and help you achieve an accurate sharpen.
0:52 – 1:05	Ideally, secure your chainsaw in a workshop vise. If you're on site, use a stump vise or similar.
1:05 – 1:11	First, find the cutting tooth that's the most damaged or smallest and begin with that one.
1:11 – 1:14	Use calipers to ensure accuracy.
1:14 – 1:19	Once you have filed that tooth, all the cutters should then be filed back to that length.
1:19 – 1:31	The reason for this is to ensure a straight even cut. Reduce vibration, reduce kickback and reduce wear on the rest of the chain. Guide bar and sprocket.
1:31 – 1:36	Mark where you start with a pen or similar to help you keep track of where you are.
1:36 – 1:43	You can also just look and see that once you have sharpened the tooth, it will have a fresh shine to
1:43 – 1:51	So let's begin by placing the file in file. Guide into the gullet of the tooth.
1:51 – 1:57	Ensure the line runs parallel with the chain.
1:57 – 2:08	Push forward, putting pressure down and back towards the tooth.
2:08 – 2:15	Make sure you always file into and through the working corner.
2:15 – 2:18	Do not file or drag the file back against the tooth.
2:19 – 2:23	This blunts the
2:23 – 2:29	Try to keep the file level and keep the strokes consistent.
2:29 – 2:38	Be careful not to angle the file upwards, as you can inadvertently file the tie straps.
2:38 – 2:42	Don't file past the witness
2:42 – 2:50	There is potential for the teeth to snap off and damage the chain.
2:50 – 2:57	You can either file all the teeth on one side first, then turn the saw around to do the other side.
2:57 – 3:11	Or you can switch between hands, filing each cutter as you go. You might only need a couple of strokes on each cutter. The aim is to get consistency throughout the chain.
3:11 – 3:25	try to keep a good side plate angle. You want to achieve a nice hooked shape like a crest of a
3:25 – 3:33	If it's too hooked. This can lead to a more aggressive cut, but it will dull faster
3:33 – 3:43	it's too vertical or laid back. It will struggle to cut anything. The side plate angle is arguably more important than the top plate angle.
3:43 – 3:48	Once all the teeth have been sharpened evenly, we need to look at the depth gauges.
3:48 – 3:55	The depth gauges are vital as they determine how much of the working corner and cutter are allowed to sever the timber.
3:55 – 3:59	This regulates the size of chip produced.
3:59 – 4:04	Take a depth gauge setting tool of your choice
4:04 – 4:07	and place on the chain.
4:07 – 4:14	If the depth gauge stands proud, it needs to be filed.
4:14 – 4:20	Tilt the guide at an angle. This also protects the working corner you've just sharpened.
4:20 – 4:32	Push the flat file over the depth gauge a few times until it's level with the guide, and round off the edge to ensure a smooth cut.
4:32 – 4:41	Do not file over the guide, as this will gradually lower it and therefore lower the height of the depth gauge.

4:41 – 4:59	If you use a blunt, badly damaged or poorly sharpened chain, you will encounter several problems such as overheating, decreased cutting efficiency, increased kickback and vibration, uneven cuts, and increased wear on the chain, guide bar, and sprocket.
4:59 – 5:05	So keep your chain razor sharp and you will cut safer and happier,
5:05 – 5:08	This concludes the video on how to sharpen your chain.

Module 25: Kickback		LO6 · AC 6.3
Timecode	Transcript	
0:01 – 0:22	In this video, we'll be covering kickback, what it is, how it happens, and what steps you can take to avoid or reduce the risk of it happening. So what's kickback? Kickback is the violent, abrupt movement of the chainsaw, usually back towards the user when the chain moves around the top quadrant of the guide bar and comes into contact with timber.	
0:22 – 0:35	The chain continues to cut. At this point, it's still a pushing chain, so pushes the saw in an upwards arc and back, kicking the chainsaw towards the operator.	
0:35 – 0:48	Pinch kickback is similar. The chain continues to cut, but pushes and rolls the chainsaw out of the cut into the operator to avoid or reduce the risk of kickback. There are several things you can do.	
0:48 – 0:54	First and foremost is body position. Keep your head and neck out the way.	
0:54 – 1:00	See the video on work positioning for a more in-depth guide.	
1:00 – 1:04	Avoid using the upper quadrant kickback zone on the guide bar.	
1:04 – 1:13	Keep your chain sharpened accurately. Choose kickback reducing chain features such as ramped depth gauges,	
1:13 – 1:18	bumper drive links	
1:18 – 1:22	bumper tie straps,	
1:22 – 1:25	keep your chainsaw in a good working order.	
1:25 – 1:34	Good habits are worth forming straight from the start. It literally might save your life.	
1:34 – 1:36	This concludes the video on kickback.	

Module 26: Work Positioning		LO6 · AC 6.1
Timecode	Transcript	
0:00 – 0:05	In this video, we'll explain the importance of good work positioning.	
0:05 – 0:16	A good work position will keep you safe. Put less strain on your body. Keeping injuries to a minimum. The immediate hazards of a poor work position are cutting yourself and kickback,	
0:17 – 0:22	while the longer term hazards are backache and body pain issues.	
0:22 – 0:24	For the correct work position.	
0:24 – 0:32	Position the chainsaw on your right hand side with your feet pointing directly in front of you, in the same direction as the chainsaw.	
0:32 – 0:45	This should ensure whatever direction your feet point in your sore should follow. Keeping your head and neck out the way your left thumb should always be wrapped around the handle to give you a good grip.	
0:45 – 0:49	Don't rest your thumb on top.	
0:50 – 0:55	Hold the chainsaw close to yourself and use your body and legs to support some of its weight.	
0:55 – 1:10	Aim to keep a straight back if the timber is low or on the floor. Crouch down and use one knee to support yourself, but avoid kneeling on both knees as this can impair your escape. If the timber rolls or moves towards you,	
1:11 – 1:18	head position is vital. Do not position your head over the guide bar when cutting.	
1:18 – 1:22	If you get kickback, it's going straight into your head or neck.	
1:23 – 1:31	If you can see the top of the chain or the outside of the guide bar, your head is in the wrong position.	
1:31 – 1:38	If you can see the inside of the guide bar, your head is in the correct position.	
1:38 – 1:44	Make positioning a high priority from the start and you'll form good habits that will last a lifetime.	

1:44 – 1:47	This concludes the video on work positioning.
-------------	---

Module 27: Pre-Start Checks LO5 · AC 5.2

Timecode	Transcript
0:01 – 0:04	In this video we'll cover the pre start checks.
0:04 – 0:12	Before you start and use your chainsaw. A number of checks need to be done to make sure the chainsaw is safe and working correctly.
0:12 – 0:19	First off, make sure your chainsaw is topped up with the appropriately mixed fuel.
0:19 – 0:25	Remember, always use a good quality two stroke oil and the ratio is 50 to 1.
0:25 – 0:27	So in a standard five liter fuel can
0:27 – 0:34	You will need 100ml of two stroke oil.
0:34 – 0:37	Always top up the chain oil when refueling.
0:37 – 0:38	Every time.
0:38 – 0:40	No exceptions.
0:41 – 0:47	If you're using an electric chainsaw, make sure the battery is in good condition and fully charged.
0:47 – 0:52	all safety features are present and there are no obvious defects.
0:52 – 0:55	The chain is correctly tensioned
0:56 – 1:00	and the nuts clamping the side cover are tight.
1:00 – 1:06	With those initial checks done, we can begin the process of starting the chainsaw.
1:06 – 1:09	This concludes the video on the pre start checks.

Module 28: Starting The Chainsaw LO5 · AC 5.1

Timecode	Transcript
0:00 – 0:15	This video will show you how to start your chainsaw when the engine is cold or warm. There are only a few starting procedures for all chainsaws, but check with your manufacturers. Make and model. Just to be on the safe side.
0:15 – 0:24	Firstly ensure you have completed the pre start checks. Now let's prepare the area and yourself for starting the chainsaw.
0:24 – 0:32	Start by placing your chainsaw on relatively flat ground and clear any debris away from the chain and bar.
0:32 – 0:52	Kneel down and put your right foot on the back plate with your left hand on the handle and apply the chain brake. Keep the source snug up to your left knee. This will help keep the source stable when you pull it over. If your chainsaw has a decompression valve, now is the time to push it in. This makes pulling the starter handle easier.
0:52 – 1:00	If your saw has got a primer bubble, press it 5 to 6 times to purge any air within the fuel system.
1:00 – 1:03	Let's cover the first starting sequence.
1:03 – 1:06	The first starting sequence is full choke,
1:06 – 1:08	half revs,
1:08 – 1:11	then idle.
1:11 – 1:22	It might have an auto return stop switch. This means that when you press the switch, it turns the saw off, but automatically returns the switch to the on position.
1:23 – 1:42	Pull the switch out and up to the full choke position. Hold the starter handle tightly and pull slowly until you feel some resistance. Pull firmly and sharply directly upwards. Do this as many times as necessary until you hear the sore cough or fire up.
1:42 – 1:43	Did you hear
1:43 – 1:54	If you missed this fire and keep pulling, you can flood the engine. So let's hear that again.
1:54 – 2:02	Once you hear this, push the switch down to the half revs position. Pull the starter handle again until the saw begins to run.

2:02 – 2:16	Leave the chainsaw for 1 to 2 seconds and then press the trigger. This will cancel the half rev setting and bring the revs down to an idle. Your saw should be happily idling and is now warm.
2:16 – 2:19	Now let's cover the second sequence.
2:19 – 2:21	The second sequence is choke,
2:21 – 2:25	then idle.
2:25 – 2:32	Hold down the trigger and move the switch down to the choke position. Keep your thumb on the switch as you let go of the trigger.
2:32 – 2:35	This keeps the switch in the full choke position.
2:35 – 2:49	Pull the starter handle until the saw fires up and begins to run. Leave the chainsaw for 1 to 2 seconds and press the trigger. Your saw should be idling and now be warm.
2:49 – 2:55	The third sequence is a primer only. You should have a symbol like this.
2:55 – 3:00	It will also have an auto return on off switch.
3:00 – 3:14	Press the primer bubble 5 to 6 times as necessary, and pull the starter handle until the saw fires up and starts to idle.
3:14 – 3:20	Once your chainsaw is warm, there's no need to repeat the cold start method.
3:20 – 3:24	We can perform a warm start instead.
3:24 – 3:30	Make sure the switch is in the on position
3:30 – 3:38	and the chain brake is applied. Brace the chainsaw securely between your legs and pull the star to handle across your body straight upwards.
3:38 – 3:49	If it doesn't start, the saw may be still a little cold and you may have to repeat the cold start sequence again.
3:49 – 3:55	Under no circumstances should you attempt to drop start the chainsaw.
3:55 – 4:00	This is a bad habit and uncontrolled movements can lead to injuries.
4:01 – 4:09	If you have a battery operated chainsaw, make sure the battery is in good working order, fully charged and slotted securely in place.
4:09 – 4:15	Engage the chain brake and turn the saw
4:15 – 4:20	Now that your chainsaw is warm and ready, we can complete the pre used checks in the next video.
4:20 – 4:23	This concludes the video on how to start your chainsaw.

Module 29: Pre-Use Checks		LO5 · AC 5.2
Timecode	Transcript	
0:01 – 0:06	In this video we'll go over the pre use checks.	
0:06 – 0:11	Once your chainsaw has been started and is warm, you must complete some pre use checks.	
0:11 – 0:26	Firstly, check the chain break is working. Disengage the chain break with your left hand fingers, keeping your thumb around the handle to make disengaging the chain break easier. Rest the guide bar on a piece of timber or on your upper thigh.	
0:26 – 0:30	This will help with fatigue in your arms.	
0:30 – 0:49	Once disengaged, squeeze the throttle and bring the sword to full revs. Let go of the throttle and half a second later. Apply the chain brake. Applying the chain brake is done by rotating the left hand forwards and straightening the right arm, forcing the back handle down.	
0:49 – 0:57	Do not rep the chainsaw for long periods whilst the chain brake is engaged.	
0:57 – 1:01	This will burn out your clutch and chain brake band.	
1:02 – 1:06	Secondly, check that the chain is being oiled.	
1:06 – 1:28	Release the chain brake and hover the nose of the guide bar over a piece of timber or stump, or something similar that will not blunt your chainsaw if mistakenly touched. Squeeze the throttle and bring to full revs over the timber. After a few more seconds, you should be able to see a faint line of oil that has been flung from the end of the bar.	
1:28 – 1:33	If you can't see any oil, some maintenance will be needed.	

1:18 – 1:22	This concludes the section on cutting timber under tension and compression.
-------------	---

Module 32: Releasing A Trapped Chainsaw		LO6 · AC 6.2
Timecode	Transcript	
0:01 – 0:06	This video is about how to release and remove a trapped or pinched chainsaw.	
0:06 – 0:15	Misjudging the tension and compression can lead to your chainsaw being trapped or pinched within the cut. Don't panic. It can be fairly common	
0:15 – 0:23	if your sword does get stuck. The first thing you need to do is turn the saw off. Make sure the chain brake is off as you may be able to roll the chainsaw out.	
0:23 – 0:25	The cut.	
0:25 – 0:30	Do not try and pull the chainsaw out with the saw still running.	
0:30 – 0:39	If the sword does come free while you're doing this, you can inadvertently rev the sword, pulling it into yourself and stumble backwards.	
0:40 – 0:47	If rolling the saw out the cut doesn't work. Try and lift or push the timber up or down at one end to release the saw.	
0:47 – 1:03	If it's bigger timber, you can drive wedges into the curve to open up the cut. This should release the saw and allow for removal. If the saw is still stuck and the timber is too heavy to lift or move. You will need another chainsaw to cut you free	
1:03 – 1:09	when using another chainsaw. Cut a minimum of 300mm or 1ft away from the trap saw.	
1:09 – 1:15	Or preferably, you cut a product's length away so you don't waste any timber.	
1:16 – 1:19	This concludes the video on removing a trapped or pinched chainsaw.	

Module 33: Bore Cutting		LO6 · AC 6.3
Timecode	Transcript	
0:00 – 0:04	This video is about performing a bore cut.	
0:04 – 0:13	The bore cut is used to cut timber when it cannot be easily accessed from one side. The most common scenario is when the timber is lying close to the floor	
0:13 – 0:16	and the compression is on top,	
0:16 – 0:20	cutting the tension without a bowl cut. We put your saw in the dirt	
0:21 – 0:25	and blunt your chain.	
0:25 – 0:30	Let's do it properly.	
0:30 – 0:40	The first method is to use the pushing chain and the top of the guide bar to make the ball. This method is useful when the timber is low and small.	
0:40 – 0:43	Start by making the compression cut on top	
0:43 – 0:48	20 to 30% of the timber's diameter.	
0:48 – 0:53	Begin the ball below the compression cut and use the top of the guide bar.	
0:53 – 1:06	The guide bar needs to be almost vertical to avoid any kickback. Cut at full revs and once the nose of the guide bar is in, smoothly, move the guide bar into the horizontal position and push all the way through the timber.	
1:06 – 1:09	Don't cut upwards to meet your compression. Cut	
1:09 – 1:16	a strip of holding timber needs to be left uncut.	
1:16 – 1:33	Once you're all the way through, cut downwards to release the tension. Be careful not to exit the cut quickly and dig into the floor. Once the tension has been removed, move the saw up and cut the holding strip of timber.	
1:33 – 1:40	The second method is to use the pulling chain and the bottom of the guide bar to make the ball	
1:40 – 1:41	again.	
1:41 – 1:52	Start by making the compression cut on top 20 to 30% of the timbers diameter. Begin the ball below the compression cut and use the bottom of the guide bar. Nose.	
1:52 – 1:56	Keep the guide bar nose high to avoid any kickback.	

1:56 – 2:07	Cut at full revs and once the nose is in, smoothly move the guide bar into the horizontal position and bore through the timber and out the other side.
2:07 – 2:12	When you're through, cut downwards to release the tension.
2:12 – 2:19	Once the tension has been removed, move the saw up and cut the remaining strip of timber.
2:19 – 2:21	The more you practice,
2:21 – 2:24	smoother you'll get.
2:24 – 2:26	This concludes the video on boar cutting.

Module 34: Oversized & Tensioned Timber		LO6 · AC 6.4
Timecode	Transcript	
0:00 – 0:06	In this video, we'll demonstrate how to cut oversize and highly tension timber	
0:07 – 0:13	for cutting oversize timber The easiest way is to get a larger guide bar. If your saw allows.	
0:13 – 0:17	If that's not an option, you can use a reduction cut.	
0:17 – 0:30	Position the chainsaw over the top of the log and cut the opposite side. Now bring the saw over the top and continue cutting down to finish the cut.	
0:30 – 0:37	You can also cut the timber from both sides	
0:37 – 0:41	or cut one side.	
0:41 – 0:44	Roll the timber	
0:44 – 0:47	and finish the cut.	
0:47 – 1:05	Now onto highly tensioned timber. Cutting under extreme tension can be quite hazardous as the timber can split and spring towards you or trap your chainsaw. Make sure you stand on the compression side to minimise the risk of these hazards.	
1:05 – 1:18	For smaller timber, you can use the toast rack method. This involves making a series of small cuts on the compression side,	
1:18 – 1:26	and once the tension has eased sufficiently, then cut through the tension	
1:26 – 1:38	for larger timber. You can cut a continually widening V shape. This should safely relieve the tension.	
1:38 – 1:42	This concludes the video on how to cut oversize and highly tension timber.	

Module 35: Stacking		LO6 · AC 6.4
Timecode	Transcript	
0:01 – 0:04	This video is about lifting and stacking timber	
0:04 – 0:14	back injuries from using chainsaws and lifting and moving. Timber can last a lifetime. So if you can let the machine do the work for you,	
0:14 – 0:21	if you have to do the work manually, keep the saw close to your body. Use your legs, not your back.	
0:21 – 0:22	And never overreach.	
0:23 – 0:25	Only lift what you can safely manage.	
0:25 – 0:47	Otherwise, drag, roll or use lifting aids like timber tongs or hooka runes. When stacking timber, keep the stacks waist height one meter or lower. Stack logs by their species and wedge them securely if needed. Work smart lift safe and keep your back strong for the long run. This concludes the video on lifting and stacking timber.	

Module 36: Additional Cuts		LO6 · AC 6.4
Timecode	Transcript	
0:00 – 0:05	In this video, we'll be covering some useful additional cuts.	
0:05 – 0:15	The first one we'll cover is the step cut. The step cut is useful when the tension is on top and there's risk of the chainsaw getting taken with the timber.	
0:15 – 0:26	To perform a step cut. Make the first cut underneath. On the compression side of the timber. Then make the second cut on the tension side inboard of the first cut	

0:26 – 0:31	and overlap the cuts.
0:31 – 0:39	Inboard is the side of least movement. So back towards where the timber is most static.
0:39 – 0:46	Don't position the tension cut outboard of the first cut.
0:46 – 0:56	You will create a small shelf which is likely to catch the chain and take the chainsaw with the timber
0:56 – 1:07	done on the inboard side. The small shelf will leave your chainsaw safely sitting there and the timber should fall away.
1:07 – 1:11	Another cut to utilise is the box cut.
1:11 – 1:18	For demonstration purposes. The cut has been done on the end of the log. To give you a better understanding of the mechanics and what to achieve.
1:18 – 1:24	Firstly, make the compression cut underneath.
1:24 – 1:29	Then bring the chainsaw over the opposite side and make a cut.
1:29 – 1:33	Then the top.
1:33 – 1:35	Then the near side.
1:35 – 1:48	All cuts should be roughly 20 to 30% of the timber's diameter. This should leave a square or box of uncut timber in the middle. Once this has been achieved, do the final cut on the tension side.
1:49 – 2:04	If you think there is a risk of the chainsaw being taken with the timber, just make the final cut inboard of the box. This acts like a step cut and keeps your chainsaw from being taken with the timber.
2:04 – 2:06	This concludes the video on additional cuts.

SECTION 7

Assessment Bank

151 Multiple-Choice Questions · Summative Examination Pool

How the exam works

The summative examination draws a randomised 45 questions from this 151-question bank. Questions are mapped to their Learning Outcome (LO) and Assessment Criterion (AC). Learners must achieve 80% or above to pass. Correct answers are highlighted in orange with ' ' .

Q1

LO1 · AC 1.4

What is the PRIMARY purpose of chainsaw protective trousers?

- A. Provide warmth in cold conditions
- B. Stop or slow a chainsaw chain on contact with the leg ' '**
- C. Improve operator visibility
- D. Protect against thorns and branches only

Q2

LO1 · AC 1.4

What is the FIRST action to take when a chainsaw operator sustains a deep cut?

- A. Apply direct pressure to control bleeding and call 999 ' '**
- B. Remove the chainsaw chain from the wound immediately
- C. Give the casualty food and water
- D. Continue working and treat the injury later

Q3

LO1 · AC 1.4

Which standard governs chainsaw protective clothing in the UK?

- A. BS EN 381 ' '**
- B. BS EN 397
- C. BS EN 471
- D. BS EN 166

Q4

LO1 · AC 1.4

How frequently must chainsaw PPE be inspected for damage?

- A. Once per year
- B. Once per month
- C. Before every use ' '**
- D. Only when it visibly looks worn

Q5

LO2 · AC 2.1

What is the FIRST step in the 5-step risk assessment process?

- A. Record your findings
- B. Identify the hazards ' '**
- C. Evaluate the risks
- D. Implement control measures

Q6

LO2 · AC 2.1

Under the Management of Health and Safety at Work Regulations 1999, who must carry out a risk assessment?

- A. The Health and Safety Executive
- B. The local authority
- C. Employers and self-employed persons ' '**
- D. Only qualified safety officers

Q7

LO2 · AC 2.1

A risk assessment must be reviewed when:

- A. A new calendar year begins
- B. Working conditions change significantly ' '**
- C. A manager verbally requests it
- D. Staff turnover occurs

Q8

LO2 · AC 2.1

In a risk assessment, what does 'risk' specifically mean?

- A. The likelihood that harm will occur from a hazard
- B. The physical object that could cause harm
- C. The financial cost of an accident
- D. The legal penalties for non-compliance

Q9

LO2 · AC 2.2

Before starting chainsaw work, what emergency information must be established?

- A. The nearest fuel station
B. The nearest hardware store
C. The site manager's home address
D. The location of emergency services and a means to contact them

Q10

LO2 · AC 2.2

What is the recommended approach for lone workers operating a chainsaw?

- A. Check in every 4 hours
B. Check in every 8 hours
C. Check in at regular agreed intervals (e.g. every 30 minutes) *
D. Check in once per day

Q11

LO2 · AC 2.2

An emergency action card should include:

- C. Site location, grid reference, and emergency service numbers

Q12

LO2 · AC 2.2

What does 'point of work risk assessment' mean?

- A. An assessment carried out at the actual working location before starting work '**
B. An assessment completed in the office before site arrival
C. A financial assessment of the job
D. An assessment completed after work is finished

Q13

LO1 · AC 1.1, AC 1.2

Which legislation requires work equipment to be safe and suitable for its intended use?

- A. Health and Safety at Work Act 1974
- B. Workplace (Health, Safety and Welfare) Regulations 1992
- C. Provision and Use of Work Equipment Regulations 1998 (PUWER) '1**
- D. Manual Handling Operations Regulations 1992

Q14

LO1 · AC 1.1, AC 1.2

Who is the primary enforcing authority for health and safety in UK forestry and arboriculture?

- D. The Health and Safety Executive (HSE)**

Q15

LO1 · AC 1.1, AC 1.2

The Health and Safety at Work Act 1974 places a duty on employees to:

- A. Take reasonable care for their own and others' health and safety**

Q16

LO1 · AC 1.1, AC 1.2

PUWER stands for:

- A. Personal Use of Work Equipment Regulations
- B. Protection and Upkeep of Work Equipment Regulations
- C. Provision and Use of Work Equipment Regulations ' '**
- D. Prevention of Unsafe Work Equipment Risks

Q17

LO3 · AC 3.3

What is the primary function of the chain brake?

- A. Stop the chain rotating rapidly in the event of kickback ' '**
- B. Sharpen the chain automatically
- C. Prevent the engine from starting accidentally
- D. Control engine speed at idle

Q18

LO3 · AC 3.3

The chain catcher on a chainsaw is designed to:

- A. Collect sawdust
- B. Hold the chain during sharpening
- C. Reduce chain noise
- D. Protect the operator if the chain breaks or derails ' '**

Q19

LO3 · AC 3.3

What is the purpose of the anti-vibration (AV) system on a chainsaw?

- A. Reduce engine noise emissions
- B. Prevent chain derailment
- C. Reduce vibration transmitted from the saw to the operator's hands ' '**
- D. Improve fuel efficiency

Q20

LO3 · AC 3.3

The front hand guard on a chainsaw:

- A. Protects the operator's hand from the chain and activates the chain brake ' '**
- B. Acts as a handhold during transport
- C. Supports the weight of the saw
- D. Prevents fuel from splashing the operator

Q21

LO4 · AC 4.1

Why is it critical to clean or replace a blocked chainsaw air filter?

- A. A blocked filter makes the saw louder
- B. A blocked filter improves chain speed
- C. A blocked filter has no effect on performance
- D. A blocked filter causes a rich mixture, poor performance and overheating ' '**

Q22

LO4 · AC 4.1

How should a foam air filter be cleaned?

- A. Washed in warm soapy water, dried fully before re-fitting ' '**
- B. Washed in petrol and left to dry
- C. Scraped clean with a metal tool
- D. Replaced immediately without attempting to clean

Q23

LO4 · AC 4.1

When inspecting the air filter, you should also check:

- A. The bar sprocket nose
- B. The spark plug gap only
- C. The filter housing and gasket for damage or leaks ' '**
- D. The chain tension only

Q24

LO4 · AC 4.1

Operating with a damaged or missing air filter will most likely cause:

- A. Improved engine performance
- B. Lower chain speed only
- C. Reduced vibration only
- D. Increased engine wear due to ingested debris '**

Q25

LO4 · AC 4.1

What should you assess when inspecting a chainsaw spark plug?

- A. Electrode condition, gap, and look for fouling or damage '**
- B. Replace it after every use regardless of condition
- C. Only inspect if the saw fails to start
- D. Clean it with water and refit

Q26

LO4 · AC 4.1

A black, sooty spark plug electrode typically indicates:

- A. A lean fuel mixture (too much air)
- B. A rich fuel mixture or excess oil in the combustion chamber '**
- C. Correct engine operation
- D. Severe overheating

Q27

LO4 · AC 4.1

What tool is needed to correctly set the spark plug electrode gap?

- A. A torque wrench
- B. A vernier calliper
- C. A multimeter
- D. A feeler gauge '**

Q28

LO4 · AC 4.1

What is the risk of overtightening a spark plug during installation?

- A. The plug may fall out during operation
- B. The engine will run too rich
- C. The cylinder head threads can be stripped or the plug body cracked '**
- D. The chain will derail

Q29

LO4 · AC 4.1

Why must cooling fins on a chainsaw cylinder be kept clean?

- A. To improve the appearance of the saw
- B. To reduce chain noise
- C. To improve fuel efficiency only
- D. To allow adequate airflow and prevent engine overheating '**

Q30

LO4 · AC 4.1

What should be used to clean clogged cooling fins?

- A. A soft brush or plastic tool to dislodge debris without damaging fins '**
- B. A metal scraper
- C. High-pressure water directly on the engine
- D. Petrol sprayed directly into the fins

Q31

LO4 · AC 4.1

Where does cooling air typically enter most chainsaw designs?

- A. Through the exhaust port
- B. Through the chain tensioner slot
- C. Through the flywheel and air intake shroud '**
- D. Through the bar mounting holes

Q32

LO4 · AC 4.1

Operating with blocked cooling fins can result in:

- A. Better fuel economy
- B. Increased chain speed
- C. Reduced vibration levels
- D. Piston seizure and major engine damage '**

Q33

LO4 · AC 4.1

What hazard does a blocked spark arrestor in the exhaust present?

- A. Increased fuel consumption only
- B. Reduced chain speed
- C. Fire risk from ejected hot carbon particles '**
- D. Loss of chain lubrication

Q34

LO4 · AC 4.1

Signs of a failing chainsaw exhaust system include:

- A. Quieter engine operation
- B. Improved fuel economy
- C. Reduced chain vibration
- D. Exhaust leaks causing burn risk and increased operator noise exposure '**

Q35

LO4 · AC 4.1

When should the spark arrestor be cleaned or replaced?

- A. Per the manufacturer's schedule or when visibly blocked '**
- B. It is a sealed unit and never serviced
- C. Only when the saw fails to start
- D. After every single use

Q36

LO4 · AC 4.1

Chainsaw exhaust fumes are particularly dangerous in enclosed or confined spaces because:

- A. They are brightly coloured and have a strong smell
- B. They cause skin irritation only
- C. Carbon monoxide is colourless and odourless and can reach lethal concentrations '**
- D. They corrode metal parts rapidly

Q37

LO4 · AC 4.1, AC 3.1

What is the purpose of the chainsaw fuel filter?

- A. To mix fuel and bar oil
- B. To store reserve fuel
- C. To heat the fuel before combustion
- D. To prevent debris entering the fuel system and damaging the carburettor '**

Q38

LO4 · AC 4.1, AC 3.1

Bar and chain oil should have what key property to be effective?

- A. Sufficient tackiness to adhere to the chain and bar during operation '**
- B. Be very thin to flow freely at all temperatures
- C. Be identical in specification to engine oil
- D. Be water-based for environmental compliance

Q39

LO4 · AC 4.1, AC 3.1

A partially blocked fuel filter will typically cause:

- A. Improved fuel economy
- B. Overfuelling and a rich mixture
- C. Fuel starvation, causing the engine to cut out under load '**
- D. No noticeable change in performance

Q40

LO4 · AC 4.1, AC 3.1

How often should chainsaw fuel and bar oil filters be inspected?

- A. Only during the annual service
- B. Only when a fuel problem is noticed
- C. Every 1,000 hours of use

D. Per the manufacturer's maintenance schedule '

Q41

LO4 · AC 4.1

What must be confirmed before starting a chainsaw using the recoil starter?

- A. Chain brake is engaged, saw is secure, and area is clear of people '**
- B. Chain is loose for easy starting
- C. Throttle is fully open
- D. Bar oil reservoir is empty

Q42

LO4 · AC 4.1

If the recoil starter cord does not retract after pulling, this most likely indicates:

- A. The engine is flooded
- B. A broken or damaged recoil spring '**
- C. The chain brake is disengaged
- D. Low fuel level

Q43

LO4 · AC 4.1

What is the correct ground-starting position for a chainsaw?

- A. Hold the saw between your knees with the bar pointing forward
- B. Balance the saw on the chain brake guard
- C. Place the rear handle under your right foot, hold the front handle, bar clear of the ground '**
- D. Hold the saw by the bar with one hand

Q44

LO4 · AC 4.1

The chain brake must be engaged during the starting procedure:

- A. Only after the engine has warmed up
- B. Never — it prevents the engine from starting
- C. Only when starting on steep slopes
- D. At all times throughout the starting procedure '**

Q45

LO4 · AC 4.1

When does the centrifugal clutch on a chainsaw typically engage the chain?

- A. At idle speed only
- B. Immediately on engine start
- C. At a higher RPM as engine speed increases above idle '**
- D. Only when the throttle trigger is fully released

Q46

LO4 · AC 4.1

A worn clutch drum will most commonly result in:

- A. The chain running at idle — a serious safety hazard '**
- B. Improved chain engagement at lower speeds
- C. Reduced fuel consumption
- D. Increased cutting speed

Q47

LO4 · AC 4.1

What is typically required to safely remove a chainsaw clutch assembly?

- A. A standard socket set only
- B. Mole grips on the flywheel
- C. Only a large screwdriver
- D. A clutch removal tool with the correct thread direction and a piston stop '**

Q48

LO4 · AC 4.1

The clutch springs are responsible for:

- A. Holding the clutch pads retracted at idle so the chain does not move '
- B. Returning the throttle trigger to idle position
- C. Tensioning the guide bar
- D. Driving the oil pump

Q49

LO4 · AC 4.3

Why must a worn drive sprocket be replaced?

- A. To improve the appearance of the saw
- B. A worn sprocket reduces vibration
- C. Worn teeth improve chain engagement
- D. To decrease wear on the chain and guidebar '

Q50

LO4 · AC 4.3

When fitting a new chain, what else should always be inspected for wear?

- A. The drive sprocket, as chain and sprocket wear together '
- B. The air filter
- C. The spark plug
- D. The fuel filter

Q51

LO4 · AC 4.3

What is the maximum recommended sprocket wear before it must be replaced?

- A. 0.1mm
- B. 1.0mm
- C. 0.5mm or as specified by the manufacturer '
- D. 2.0mm

Q52

LO4 · AC 4.3

A rim sprocket design allows:

- A. Greater cutting accuracy
- B. Higher idle speeds
- C. Direct drive to the chain without a clutch
- D. The rim sprocket can be replaced without replacing the clutch drum as well '

Q53

LO3 · AC 3.3

The inertia-activated chain brake operates by:

- A. The operator pressing a dedicated button
- B. The front hand guard being pressed by the operator's wrist
- C. A flyweight mechanism detecting rapid saw body rotation and triggering automatically '
- D. Releasing the throttle trigger

Q54

LO3 · AC 3.3

How should the chain brake be tested in the workshop environment?

- A. Rev to full throttle and deliberately trigger kickback
- B. Check it in the workshop with the bar removed
- C. It does not require testing before use
- D. Reassemble the saw and activate the chain brake and try to spin the chain manually '

Q55

LO3 · AC 3.3

If the chain continues rotating when the chain brake is activated, you must:

- A. Stop work immediately — do not use the saw until the brake is repaired '
- B. Continue working but exercise extra care
- C. Increase the engine idle speed
- D. Tighten the guide bar nuts

Q56

LO3 · AC 3.3

What should be checked when maintaining the chain brake?

- A. Oil the brake band lightly after each use to prevent corrosion
- B. The brake band must be clean and free of oil or grease — replace if worn, glazed, or below minimum thickness '**
- C. The brake spring should be removed during long-term storage
- D. Tighten the front hand guard pivot to increase braking force

Q57

LO4 · AC 4.3

Why should the guide bar be regularly rotated 180 degrees?

- A. To improve chain speed
- B. To change the cutting direction
- C. To allow even wear on both sides of the bar rail '**
- D. To clear the bar oil feed holes

Q58

LO4 · AC 4.3

What should be checked in the guide bar groove during inspection?

- A. Fuel tank level
- B. Bar length only
- C. Drive link count
- D. Groove depth, rail burrs, and bar oil holes for blockage '**

Q59

LO4 · AC 4.3

A guide bar with a pinched (closed or damaged) rail groove causes:

- A. Reduced chain lubrication and premature bar and chain wear '**
- B. Improved cutting accuracy
- C. Faster chain speed
- D. Reduced vibration

Q60

LO4 · AC 4.3

Bar oil feed holes must be kept clear to ensure:

- A. Correct chain tension is maintained
- B. Correct chain brake function
- C. Adequate lubrication of the chain and bar throughout cutting '**
- D. Engine cooling airflow

Q61

LO4 · AC 4.4

Drive links on a chainsaw chain serve to:

- A. Cut through the wood
- B. Reduce vibration transmission
- C. Measure the chain length
- D. Engage with the drive sprocket and guide bar groove to drive and guide the chain '**

Q62

LO4 · AC 4.4

What is a semi-chisel chain profile good for?

- A. Cutting timber under tension
- B. Cutting mostly softwood with increased kickback
- C. Cutting mostly hardwood with reduced kickback '**
- D. Cutting metal and other foreign objects

Q63

LO4 · AC 4.4

The depth gauge (raker) on a chain cutter controls:

- A. Chain speed
- B. Chain tension
- C. The depth of cut each cutter tooth takes into the wood '**
- D. Bar lubrication flow

Q64

LO4 · AC 4.4

Which part of the chainsaw chain actually cuts the wood?

- A. The tie straps
- B. The drive links
- C. The left and right cutter teeth ' '**
- D. The bumper drive links

Q65

LO4 · AC 4.4

How is correct chain tension identified on a standard chainsaw?

- A. The chain hangs loosely below the bar
- B. The chain cannot be moved by hand at all
- C. The chain pulls slightly away from the bar but snaps back with drive links remaining in the groove ' '**
- D. Pliers are needed to move the chain along the bar

Q66

LO4 · AC 4.4

Why is operating with a chain that is too loose dangerous?

- A. It reduces fuel consumption
- B. It improves cutting performance
- C. It reduces operating noise
- D. A loose chain can derail from the bar, creating a serious injury risk ' '**

Q67

LO4 · AC 4.4

When should chain tension be checked?

- A. Before use and at regular intervals — new chains need especially frequent re-tensioning ' '**
- B. Only at the start of the working day
- C. Only when the chain appears loose
- D. Once per week only

Q68

LO4 · AC 4.4

A chain that is too tight will cause:

- A. Improved cutting speed
- B. Excessive wear on bar, chain, and sprocket, and potentially dangerous bar heating ' '**
- C. Reduced kickback risk
- D. Better lubrication distribution

Q69

LO4 · AC 4.4

How do you identify a Stihl chainsaw chain?

- A. Check the colour of the tie straps and cutter teeth
- B. Count the number of cutters and divide by two
- C. Find the number below the depth gauge and on the drive link, then cross-reference with the chain chart ' '**
- D. Look for the Stihl logo stamped on the guide bar

Q70

LO4 · AC 4.4

Where on a chainsaw chain is the identification number stamped?

- A. On the tie straps
- B. On the depth gauge only
- C. On the drive link or on the cutting tooth below the depth gauge ' '**
- D. On the cutter tooth face only

Q71

LO4 · AC 4.4

What does the chain identification number allow you to do?

- A. Set the correct chain tension
- B. Determine the bar oil viscosity
- C. Identify the pitch, gauge and which file size and filing angle to use ' '**
- D. Calculate the correct bar oil viscosity

Q72

LO4 · AC 4.4

How do you identify a Husqvarna chainsaw chain?

- A. By the number of cutting teeth on the top of the chain
- B. By the colour of the depth gauge
- C. By the chain weight in grams

D. Find the part number stamped on the drive link and cross-reference with the chain identification chart ' '

Q73

LO4 · AC 4.4

What is the pitch of a chainsaw chain?

- A. The overall length of the chain in inches
- B. The width of the drive link that fits the guide bar groove

C. The average distance between three consecutive rivets divided by two ' '

- D. The angle of the cutting tooth top plate

Q74

LO4 · AC 4.4

What is the gauge of a chainsaw chain?

- A. The number of drive links in the chain

B. The thickness of the drive link that fits into the guide bar groove ' '

- C. The distance between consecutive rivets

- D. The height of the depth gauge above the cutter

Q75

LO4 · AC 4.4

What are the two main cutter profiles used on chainsaw chains?

- A. Round and square

B. Semi-chisel and full-chisel ' '

- C. Micro and macro chisel

- D. Left and right profile only

Q76

LO4 · AC 4.4

When selecting a replacement chain, which three measurements must match the original?

- A. Bar length, engine displacement, and chain weight

- B. Chain colour, manufacturer, and number of cutters

C. Pitch, gauge, and number of drive links ' '

- D. Cutter height, depth gauge setting, and chain age

Q77

LO4 · AC 4.4

When sharpening a cutter, the file should be held at:

A. The manufacturer-specified horizontal angle (typically 25-35 degrees) ' '

- B. 90 degrees straight across the chain

- C. 45 degrees downward into the cutter

- D. Parallel to the guide bar

Q78

LO4 · AC 4.4

The depth gauge (raker) on a chain cutter:

- A. Guides the file during sharpening

- B. Measures chain tension directly

- C. Indicates when the chain needs full replacement

D. Limits how deeply each cutter bites into the wood, controlling chip thickness ' '

Q79

LO4 · AC 4.4

If depth gauges become too high relative to the cutters after repeated sharpening:

- A. The chain cuts more aggressively

- B. Chain speed increases significantly

C. The chain cuts poorly because the cutters cannot engage the wood adequately ' '

- D. There is no effect on cutting performance

Q80

LO4 · AC 4.4

After sharpening, all cutters on the chain must be:

- A. Different lengths to compensate for uneven wood hardness
- B. The same length and sharpened to the same angle for smooth, efficient cutting '**
- C. Longer on one side than the other
- D. Set to varying depth gauge heights

Q81

LO6 · AC 6.4

What is the maximum recommended height for a freestanding timber stack without additional support?

- A. 1.0 metre, as specified in the site risk assessment '**
- B. 3.0 metres
- C. There is no recommended maximum height
- D. 1.5 metres

Q82

LO6 · AC 6.4

When stacking multiple different species, what should you do?

- A. Stack them randomly to save time
- B. Arrange and separate species into their respective stacks '**
- C. Mix species to improve stability
- D. Only stack one species per day

Q83

LO6 · AC 6.4

The stacking work area should be:

- A. Uneven ground to give better grip to stacked timber
- B. As close to felling operations as possible regardless of slope
- C. On stable, clear ground away from slope run-out zones '**
- D. Directly beside any active felling work

Q84

LO6 · AC 6.4

Before approaching an existing timber stack for further work, you must:

- A. Approach quickly to check for hazards
- B. Place additional timber on top to test stability
- C. Ignore the stack and focus on other nearby work
- D. Visually assess the stack for signs of instability before approaching '**

Q85

LO6 · AC 6.1

The recommended cutting stance for cross-cutting timber is:

- A. Feet together, facing the cut directly
- B. Kneeling beside the log
- C. Balanced stance with feet shoulder-width apart, good footing, body weight evenly distributed '**
- D. Standing directly over the log

Q86

LO6 · AC 6.1

Why must the operator never stand directly in line with the bar when cross-cutting?

- A. It reduces cutting efficiency
- B. It is harder to see the cut line accurately
- C. It increases vibration exposure to the legs
- D. In the event of kickback, the bar swings towards the operator's face and upper body '**

Q87

LO6 · AC 6.1

When should the operator plan and check their escape route?

- A. Before starting any cut '**
- B. After the cut is completed
- C. Only when working on steep slopes
- D. Only when working in proximity to other people

Q88

LO6 · AC 6.1

Why do you need to keep your thumb wrapped around the handle?

- A. To reduce hand fatigue during long cuts
- B. To provide a firm secure grip '**
- C. To activate the throttle lock more easily
- D. To improve vibration dampening

Q89

LO5 · AC 5.2

Which of the following is NOT typically part of a pre-start chainsaw check?

- A. Checking fuel and bar oil levels
- B. Inspecting chain brake function
- C. Adjusting the carburettor idle speed '**
- D. Checking chain tension and sharpness

Q90

LO5 · AC 5.2

If fuel is leaking from the saw during a pre-start check, you should:

- A. Stop — do not start the saw and arrange repair before use '**
- B. Continue but check it again at lunch
- C. Use the saw quickly before the leak worsens
- D. Wipe the fuel away and proceed normally

Q91

LO5 · AC 5.2

Pre-start checks should be carried out:

- A. Once per week
- B. Only when the operator changes
- C. Before each work session and after the saw has been left unattended '**
- D. Only at the start of a new contract

Q92

LO5 · AC 5.2

During a pre-start chain inspection, you are checking for:

- A. Number of cutting teeth only
- B. Chain colour and manufacturer markings
- C. Chain speed at idle
- D. Sharpness, correct tension, drive links in groove, and no damaged links '**

Q93

LO5 · AC 5.1

Why must the chain brake be engaged before starting a chainsaw?

- A. It is not necessary to engage the brake for starting
- B. To increase compression for easier starting
- C. To warm the engine before operational use
- D. To prevent chain rotation during the starting procedure, reducing injury risk '**

Q94

LO5 · AC 5.1

The correct ground-starting position for a chainsaw is:

- A. Place rear handle under right foot, hold front handle with bar clear of ground '**
- B. Hold front handle, place saw between the knees
- C. Hang saw by the front guard and pull cord upward
- D. Start with the bar held vertically

Q95

LO5 · AC 5.1

A flooded chainsaw engine is typically caused by:

- A. Too little fuel reaching the engine
- B. A blocked air filter
- C. Excess fuel in the engine from incorrect choke or throttle use during starting '**
- D. Low engine compression

Q96

LO5 · AC 5.1

After starting a cold engine, why must the choke be returned to the run position?

- A. The choke improves fuel efficiency during warm-up
- B. Leaving the choke on causes an over-rich mixture that stalls or damages the engine '**
- C. The choke controls chain speed at idle
- D. The choke only affects the initial starting phase

Q97

LO5 · AC 5.2

The throttle trigger interlock must be confirmed to:

- A. Open the throttle smoothly when fully pressed
- B. Rev the engine to maximum speed
- C. Prevent throttle engagement unless the interlock is simultaneously depressed '**
- D. Allow the chain to move at idle speed

Q98

LO5 · AC 5.2

How is bar oil reaching the chain confirmed during a pre-use check?

- A. Confirm the bar oil reservoir is completely empty before refilling
- B. Remove the oil cap during operation to observe flow
- C. Run the chain dry for the first 10 minutes and then check
- D. Hold saw over light-coloured material and check for oil spray while revving '**

Q99

LO5 · AC 5.2

Why should you check the chain is static whilst idling?

- A. To confirm the chain brake is not engaged
- B. To ensure the chain is not moving unintentionally '**
- C. To measure idle speed accuracy
- D. To check bar oil is reaching the chain

Q100

LO5 · AC 5.2

If the chain brake does NOT stop the chain when tested during a pre-use check, you must:

- A. Continue working but remain extra vigilant
- B. Tighten the guide bar nuts and retest
- C. Take the saw out of service immediately and arrange for repair '**
- D. Engage the brake manually with your hand as a workaround

Q101

LO6 · AC 6.1

What is the difference between a pushing chain and a pulling chain?

- A. A pushing chain cuts on the upstroke; a pulling chain cuts on the downstroke
- B. The pulling chain (bottom of bar) draws the saw into the cut; the pushing chain (top of bar) pushes the saw back towards the operator '**
- C. Pushing chains are used on petrol saws; pulling chains on battery saws
- D. There is no difference — all chainsaw chains work identically

Q102

LO6 · AC 6.1

Why is it important to secure timber before cross-cutting?

- A. To improve the visual accuracy of the cut line
- B. To prevent the saw from overheating during the cut
- C. To reduce the need for bumper spike contact
- D. To ensure the timber cannot roll or move unexpectedly, which could trap the bar or endanger the operator '**

Q103

LO6 · AC 6.1

What happens if the chain contacts the ground or a foreign object during cutting?

- A. The saw engine stalls automatically as a safety measure
- B. The chain brake engages to protect the operator
- C. The chain will immediately become blunt or damaged, reducing cutting efficiency and increasing hazard '**
- D. Chain speed increases due to reduced resistance

Q104

LO6 · AC 6.1

When making a standard downward cross-cut, which part of the bar should you use?

- A. The nose of the bar for maximum reach
- B. The upper edge of the bar (pushing chain)
- C. Either edge — it makes no difference
- D. The lower edge of the bar (pulling chain) to draw the saw through the timber in a controlled manner '**

Q105

LO6 · AC 6.1, AC 6.2

When a log is in compression on its upper face (supported at both ends), the safe cutting technique is:

- A. Cut straight through from the top in a single pass
- B. Cut the compression side first (from the top), then finish the cut from below '**
- C. Cut from the end of the log inward
- D. Do not attempt to cut logs in compression

Q106

LO6 · AC 6.1, AC 6.2

When a log is in tension on its upper face (cantilevered, supported at one end), the safe technique is:

- A. Cut the top (tension side) first, then finish from below
- B. Cut straight through from the top in a single pass
- C. Cut the compression side first (from below), then finish the cut from the top '**
- D. Always cut from the side to avoid all risk

Q107

LO6 · AC 6.1, AC 6.2

If a bar becomes pinched in a compression cut, what is the likely outcome?

- A. The chain cuts faster to free itself
- B. The chain brake activates automatically
- C. The saw will safely shut down
- D. The wood grips the bar, potentially stalling the saw or causing loss of control '**

Q108

LO6 · AC 6.1, AC 6.2

Before any cross-cut, the operator must assess:

- A. Where tension and compression zones are to determine the safe cutting sequence '**
- B. The colour and species of the timber only
- C. Only whether the timber is wet or dry
- D. Only the diameter of the timber

Q109

LO6 · AC 6.2

If the chainsaw bar becomes pinched during a cut, the FIRST action is:

- A. Pull the saw repeatedly until it comes free
- B. Rev the engine to maximum to force the bar free
- C. Remove the chain from the bar while it remains pinched
- D. Turn the saw off and assess the situation '**

Q110

LO6 · AC 6.2

A safe method to release a pinched bar without power tools is:

- A. Lever the kerf open using a wedge or felling lever inserted into the cut '**
- B. Strike the log firmly with the chainsaw body
- C. Cut from a completely different direction at full speed without further assessment
- D. Stand on the log and pull the saw upward with both hands

Q111

LO6 · AC 6.2

When using a felling wedge to release a trapped bar, it should be inserted:

- A. Into the kerf on the compression side to open the cut '**
- B. Into the kerf on the tension side
- C. Using the chainsaw bar itself to drive the wedge in
- D. While the chain is still running at full speed

Q112

LO6 · AC 6.2

If the engine stalls while the bar is pinched in timber, you should:

- A. Restart the engine immediately and force the cut through
- B. Use bare hands to manually open the kerf
- C. Make the saw safe and use hand tools or a second saw to release it safely ' '**
- D. Leave the saw in the timber and continue other work

Q113

LO6 · AC 6.3

Bore cutting is performed using:

- A. The top of the bar in a controlled downward motion
- B. The full length of the bar in a rapid thrust
- C. The lower part of the bar nose, entering the wood to minimise kickback risk ' '**
- D. The chain at idle speed only

Q114

LO6 · AC 6.3

Why is bore cutting considered a higher-risk technique?

- A. It consumes significantly more fuel
- B. It requires a longer bar than standard cross-cutting
- C. It is only a risk in wet or icy conditions
- D. It uses the bar nose area where kickback is most likely to be initiated ' '**

Q115

LO6 · AC 6.3

Before performing a bore cut, the operator must:

- A. Assess wood tension and compression and confirm a clear escape route ' '**
- B. Loosen the chain for greater flexibility
- C. Set the engine to idle speed for safety
- D. Ensure the bar is shorter than 35cm

Q116

LO6 · AC 6.3

Bore cutting is most commonly used for:

- A. Routine cross-cutting of felled timber
- B. Sharpening the chain in the field
- C. Cutting timber where access is limited from one side ' '**
- D. Felling trees with a diameter greater than bar length

Q117

LO6 · AC 6.4

When timber diameter exceeds the guide bar length, the correct cross-cutting technique is:

- A. Fit a longer bar before attempting the cut
- B. Cut at an angle to reach the centre of the log
- C. Do not attempt — the timber cannot be safely cut
- D. Cut from one side, then reposition and cut from the opposite side ' '**

Q118

LO6 · AC 6.4

The key additional hazard when working with large diameter timber is:

- A. Roll hazard — large logs can roll unexpectedly during or after cutting ' '**
- B. The increased weight of the chainsaw
- C. Increased noise levels from the larger log
- D. Excessive sawdust production

Q119

LO6 · AC 6.4

Tensioned timber creates a specific risk of:

- A. The wood burning during the cutting process
- B. The chain becoming magnetised
- C. Sudden and unpredictable movement of log sections endangering the operator ' '**
- D. Fuel contamination from the timber

Q120

LO6 · AC 6.4

Before cutting oversized or tensioned timber, the operator must:

- A. Thoroughly assess the timber, identify tension/compression zones, plan all cuts, and confirm escape route ' '**
- B. Begin cutting immediately to release the tension safely
- C. Ask a bystander to watch from a safe distance
- D. Wait 10 minutes after the timber has come to rest before cutting

Q121

LO3 · AC 3.2

What is the primary advantage of a battery-powered chainsaw over a petrol chainsaw?

- A. Greater cutting power for heavy timber
- B. Longer run time without stopping
- C. Cheaper to purchase initially
- D. Zero exhaust emissions and lower noise levels, making it suitable for indoor or urban environments ' '**

Q122

LO3 · AC 3.2

How should a lithium-ion chainsaw battery be stored when not in use for an extended period?

- A. Fully discharged in a sealed plastic bag
- B. Fully charged in direct sunlight to maintain cell voltage
- C. At approximately 40–60% charge in a cool, dry place away from extreme temperatures ' '**
- D. Submerged in oil to prevent corrosion

Q123

LO3 · AC 3.2

Which of the following is a key limitation of battery-powered chainsaws compared to petrol models?

- A. They cannot be fitted with a chain brake
- B. They require more PPE than petrol chainsaws
- C. Run time is limited by battery capacity, requiring charged spare batteries for prolonged use ' '**
- D. They are heavier due to the engine block

Q124

LO3 · AC 3.2

What safety check is unique to a battery-powered chainsaw before use?

- A. Verify the battery charge level and inspect the battery and terminals for damage before fitting ' '**
- B. Check the spark plug gap
- C. Check the fuel-to-oil mix ratio
- D. Test the recoil starter mechanism

Q125

LO1 · AC 1.1

What condition is caused by prolonged exposure to hand-arm vibration from chainsaws?

- A. Carpal tunnel syndrome only
- B. Hand-Arm Vibration Syndrome (HAVS) ' '**
- C. Repetitive strain injury (RSI)
- D. Arthritis

Q126

LO1 · AC 1.1

Which of the following is an early symptom of Hand-Arm Vibration Syndrome (HAVS) that should be reported to an employer?

- A. Improved grip strength after extended use
- B. Tingling, numbness or whitening of fingers, particularly in cold conditions ' '**
- C. Mild muscle soreness in the upper back only
- D. Increased sensitivity to heat in the palms

Q127

LO1 · AC 1.1

What is the MAIN risk to hearing from chainsaw operation?

- A. Temporary hearing loss only, which always recovers
- B. Permanent noise-induced hearing loss ' '**
- C. Tinnitus that resolves after rest
- D. Pressure damage from exhaust fumes

Q128

LO1 · AC 1.1

Chainsaw exhaust fumes contain which dangerous gas that can build up in poorly ventilated areas?

- A. Carbon dioxide (CO₂)
- B. Nitrogen dioxide (NO₂)
- C. Carbon monoxide (CO)**
- D. Sulphur dioxide (SO₂)

Q129

LO1 · AC 1.2

What standard must a safety helmet worn during chainsaw work meet in the UK?

- A. BS EN 388
- B. BS EN 397**
- C. BS EN 471
- D. BS EN 166

Q130

LO1 · AC 1.2

When must hearing protection be worn during chainsaw operation?

- A. Only when working within 10 metres of other people
- B. Only on tasks lasting over 30 minutes
- C. At all times when the saw is running — chainsaw noise routinely exceeds safe exposure limits**
- D. Only when instructed by a supervisor on site

Q131

LO2 · AC 2.2

Under RIDDOR 2013, which type of accident at work must be reported to the HSE?

- A. Any bruise or scrape
- B. Over-three-day injuries only
- C. Specified injuries, dangerous occurrences, and deaths**
- D. Minor cuts requiring no medical treatment

Q132

LO2 · AC 2.2

What type of fire extinguisher is safest to use on a chainsaw fuel fire?

- A. Water
- B. CO₂ or dry powder**
- C. Foam only
- D. Wet chemical

Q133

LO2 · AC 2.2

If a chainsaw operator collapses due to suspected carbon monoxide poisoning, what is the FIRST action?

- A. Give the casualty water
- B. Move the casualty to fresh air immediately**
- C. Apply a tourniquet
- D. Administer pure oxygen from a canister

Q134

LO2 · AC 2.2

What information should be written on a site emergency card before work begins?

- A. Operator's personal insurance details
- B. Grid reference or address, nearest hospital, and emergency contact numbers**
- C. The operator's next of kin only
- D. The local authority planning reference

Q135

LO3 · AC 3.3

What does the right-hand rear guard (back hand guard) on a chainsaw protect the operator from?

- A. Fuel spray during refuelling
- B. Injury to the right hand if the chain breaks or derails from the guide bar**
- C. Exhaust heat on the throttle hand
- D. Accidental contact with the front hand guard

Q136

LO3 · AC 3.3

Which chain brake activation mechanism fires automatically during kickback WITHOUT requiring physical contact with the front hand guard?

- A. The throttle interlock mechanism
- B. The rear hand guard pressure sensor
- C. The inertia-activated brake, triggered by rapid rotation of the saw body '**
- D. The decompression valve releasing

Q137

LO3 · AC 3.3

How often should the chain brake function be checked as part of routine chainsaw operation?

- A. Once per week before the first job of the week
- B. Before every work session as part of pre-use checks '**
- C. Only after a suspected kickback incident has occurred
- D. During the annual service only

Q138

LO3 · AC 3.3

If anti-vibration (AV) mounts are found to be cracked or hardened during a chainsaw inspection, the correct action is:

- A. Continue using the saw but limit each session to 30 minutes
- B. Apply silicone sealant to the cracks and continue monitoring
- C. Do not use the saw — arrange replacement of AV mounts before further operation '**
- D. Only replace the mounts during the next scheduled annual service

Q139

LO3 · AC 3.2

What does the depth gauge (raker) on a chainsaw chain control?

- A. The angle at which the file is held during sharpening
- B. The speed at which the chain runs around the bar
- C. The depth of cut — how much material each cutter tooth removes with each pass '**
- D. The tension of the chain on the guide bar

Q140

LO3 · AC 3.2

If depth gauges are set too low on a chainsaw chain, what is the most likely effect?

- A. The chain will not cut
- B. Increased vibration and kickback risk '**
- C. The chain will slip off the bar
- D. Reduced fuel consumption

Q141

LO4 · AC 4.1

How should chain tension be checked on a chainsaw?

- A. The chain should sag visibly below the bar
- B. The chain should sit flush with the bar groove and snap back when pulled '**
- C. The chain should be tight enough that it cannot be moved by hand
- D. Chain tension should only be checked when the chain is cold

Q142

LO4 · AC 4.1

What does the drive link on a chainsaw chain do?

- A. Connects the cutting teeth to the tie straps
- B. Fits into the bar groove and is driven by the sprocket '**
- C. Provides the depth gauge function
- D. Attaches the chain to the clutch drum

Q143

LO4 · AC 4.1

Why must the chainsaw bar groove be cleaned regularly?

- A. To improve the appearance of the bar
- B. To prevent sawdust build-up that can cause the chain to jam or the bar to overheat '**
- C. To allow the chain to run faster
- D. To prevent rust forming on the bar tip

Q144

LO4 · AC 4.1

In which direction should a correctly fitted chainsaw chain travel along the top of the bar?

- A. Away from the operator (towards the bar tip) '
- B. Towards the operator
- C. Either direction is acceptable
- D. The direction depends on wood species

Q145

LO5 · AC 5.1

As part of a pre-start inspection, how should the operator confirm the bar oil tank has been filled?

- A. Remove the bar and tip the saw to check for oil flow
- B. Check the oil cap is tight — no further check required
- C. Check the oil tank level visually or with a dipstick and fill before starting '
- D. Run the saw for 30 seconds first to prime the oiling system

Q146

LO5 · AC 5.1

During a pre-start guide bar inspection, which condition would require the bar to be taken out of service?

- A. Minor surface rust on the bar rails
- B. Sawdust packed in the oil feed hole (cleaned before use)
- C. A cracked or bent bar, or rails worn beyond manufacturer specification '
- D. A bar sprocket nose making occasional clicking noises

Q147

LO5 · AC 5.2

When performing a warm start on a chainsaw that has recently been running, the choke should be:

- A. Set to half-choke to enrich the warm fuel mixture
- B. Fully closed until the engine fires on the first pull
- C. Left in the open (run) position — no choke is needed for a warm engine '
- D. Adjusted to the manufacturer's warm-start position on the carburettor

Q148

LO5 · AC 5.2

During a pre-use check, why must the operator confirm that all controls return to their neutral position when released?

- A. To confirm fuel is reaching the carburettor correctly
- B. To prevent the chain running unintentionally, reducing injury risk during operation '
- C. To check the chain brake is correctly disengaged
- D. To verify engine temperature is within normal range

Q149

LO6 · AC 6.2

A log is resting on two supports with the cut point in the middle. Where is the compression zone, and how should the cut be started?

- A. Compression at the bottom — start cutting from below and cut upward
- B. Compression at the top — start from below (tension side) so the kerf opens rather than pinches '
- C. No compression zone exists when both ends are supported
- D. Compression at the top — start from the top to cut through it first

Q150

LO6 · AC 6.2

A log is cantilevered — supported at one end only, with the cut near the free overhanging end. Where is the compression zone?

- A. At the top of the log — start from above and cut downward through the tension zone last
- B. At the bottom of the log — start from above (tension side) to avoid bar pinch from below '
- C. There is no compression in an overhanging log
- D. At the top of the log — start from below so compression holds the kerf open

Q151

LO1 · AC 1.1, AC 1.2

Under the Manual Handling Operations Regulations 1992 (MHOR), where manual handling involving a risk of injury cannot be avoided, what must an employer do?

- A. Issue all workers with a manual handling certificate
- B. Make a suitable assessment of the manual handling task and take appropriate steps to reduce the risk of injury '
- C. Prohibit manual handling activities entirely on site
- D. Ensure a written permit to lift is issued before any log is moved

SECTION 8

Supplementary Oral & Practical Mock Questions

79 Questions · Formative Practice Only — Not Formally Assessed

Important — these questions do not contribute to the pass/fail outcome

These 79 questions are oral and practical preparation aids. They do NOT form part of the summative examination, do NOT appear on the certificate, and are NOT formally assessed. Within the app, learners practise them via an AI-assisted voice or text feature that provides formative feedback.

M1

Oral / Practical · Formative Only

What are the 5 steps to risk assessment?*Key points:*

- [object Object]

M2

Oral / Practical · Formative Only

Give me at least three hazards with the site and their control measures.*Key points:*

- [object Object]

M3

Oral / Practical · Formative Only

Give me at least three hazards with the chainsaw and their control measures.*Key points:*

- [object Object]

M4

Oral / Practical · Formative Only

Give me at least three hazards with the job to be done and the control measures.*Key points:*

- [object Object]

M5

Oral / Practical · Formative Only

What emergency information do I need to know for the site?*Key points:*

- [object Object]

M6

Oral / Practical · Formative Only

What is the Health and Safety at Work Act all about?*Key points:*

- [object Object]

M7

Oral / Practical · Formative Only

Under POWER, what does it say about the equipment?*Key points:*

- [object Object]

M8

Oral / Practical · Formative Only

Who provides industry guidance for tree work?*Key points:*

- [object Object]

M9

Oral / Practical · Formative Only

Why is it important to maintain the saw to the manufacturer's specifications?*Key points:*

- [object Object]

M10

Oral / Practical · Formative Only

The manufacturer has installed safety features on the saw — run through where they are and what they do.

Key points:

- [object Object]

M11

Oral / Practical · Formative Only

If you find one of the safety features is broken or missing, what will you do?

Key points:

- [object Object]

M12

Oral / Practical · Formative Only

What are the advantages of using battery chainsaws?

Key points:

- [object Object]

M13

Oral / Practical · Formative Only

What are the disadvantages of using battery chainsaws?

Key points:

- [object Object]

M14

Oral / Practical · Formative Only

What maintenance might you do with the batteries, the saw and charger?

Key points:

- [object Object]

M15

Oral / Practical · Formative Only

Can you take your top cover off your chainsaw please.

Key points:

- [object Object]

M16

Oral / Practical · Formative Only

So what's this?

Key points:

- [object Object]

M17

Oral / Practical · Formative Only

What does it do? (air filter)

Key points:

- [object Object]

M18

Oral / Practical · Formative Only

How do you maintain it? (air filter)

Key points:

- [object Object]

M19

Oral / Practical · Formative Only

So what's this? (spark plug)

Key points:

- [object Object]

M20

Oral / Practical · Formative Only

What does it do? (spark plug)

Key points:

- [object Object]

M21	Oral / Practical · Formative Only
What are you looking for and how would you maintain it? (spark plug)	
<i>Key points:</i> • [object Object]	
M22	Oral / Practical · Formative Only
In conjunction with the flywheel, what do these cylinder fins help do?	
<i>Key points:</i> • [object Object]	
M23	Oral / Practical · Formative Only
How do you maintain the cooling system?	
<i>Key points:</i> • [object Object]	
M24	Oral / Practical · Formative Only
So what's this? (exhaust)	
<i>Key points:</i> • [object Object]	
M25	Oral / Practical · Formative Only
What does it do? (exhaust)	
<i>Key points:</i> • [object Object]	
M26	Oral / Practical · Formative Only
How do you maintain it? (exhaust)	
<i>Key points:</i> • [object Object]	
M27	Oral / Practical · Formative Only
What is in here that you need to maintain and how would you do it? (fuel tank)	
<i>Key points:</i> • [object Object]	
M28	Oral / Practical · Formative Only
What is in here that you need to maintain and how would you do it? (oil tank)	
<i>Key points:</i> • [object Object]	
M29	Oral / Practical · Formative Only
This is the recoil housing — can you take it off and de-tension the pull cord please.	
<i>Key points:</i> • [object Object]	
M30	Oral / Practical · Formative Only
Where would you expect to find damage? (recoil starter)	
<i>Key points:</i> • [object Object]	
M31	Oral / Practical · Formative Only
Now can you take off the chain and bar side cover please.	
<i>Key points:</i> • [object Object]	

M32

Oral / Practical · Formative Only

So what's this? (sprocket)*Key points:*

- [object Object]

M33

Oral / Practical · Formative Only

What does it do? (sprocket)*Key points:*

- [object Object]

M34

Oral / Practical · Formative Only

How do you check for any wear or damage? (sprocket)*Key points:*

- [object Object]

M35

Oral / Practical · Formative Only

If you need to replace the sprocket, how are you going to do that?*Key points:*

- [object Object]

M36

Oral / Practical · Formative Only

Now on to the chain brake — where would you find it on your saw?*Key points:*

- [object Object]

M37

Oral / Practical · Formative Only

What does the chain brake do?*Key points:*

- [object Object]

M38

Oral / Practical · Formative Only

How do you check the chain brake for any wear or damage?*Key points:*

- [object Object]

M39

Oral / Practical · Formative Only

This is your guidebar — what signs of wear or damage are you looking for?*Key points:*

- [object Object]

M40

Oral / Practical · Formative Only

If the bar shows excessive wear or damage, what problems might you get?*Key points:*

- [object Object]

M41

Oral / Practical · Formative Only

How are you going to maintain your bar?*Key points:*

- [object Object]

M42

Oral / Practical · Formative Only

Can you reassemble your saw please.*Key points:*

- [object Object]

M43	Oral / Practical · Formative Only
When you're ready, can you clamp your saw into a vice please. <i>Key points:</i> • [object Object]	
M44	Oral / Practical · Formative Only
How do you identify your chain? <i>Key points:</i> • [object Object]	
M45	Oral / Practical · Formative Only
What information do you need to replace your chain? <i>Key points:</i> • [object Object]	
M46	Oral / Practical · Formative Only
What cutter profile do you have on your chain? <i>Key points:</i> • [object Object]	
M47	Oral / Practical · Formative Only
What is the other main type of cutter profile? <i>Key points:</i> • [object Object]	
M48	Oral / Practical · Formative Only
And what are their different uses? <i>Key points:</i> • [object Object]	
M49	Oral / Practical · Formative Only
What top plate angle are you filing at? <i>Key points:</i> • [object Object]	
M50	Oral / Practical · Formative Only
And what file size are you using? <i>Key points:</i> • [object Object]	
M51	Oral / Practical · Formative Only
So where are you going to start sharpening? <i>Key points:</i> • [object Object]	
M52	Oral / Practical · Formative Only
So what's this? (depth gauge / raker) <i>Key points:</i> • [object Object]	
M53	Oral / Practical · Formative Only
What does it do? (depth gauge) <i>Key points:</i> • [object Object]	

M54

Oral / Practical · Formative Only

Why does the depth gauge need to be set correctly, and what's the danger in setting it too low?*Key points:*

- [object Object]

M55

Oral / Practical · Formative Only

Why do you want to keep all your cutters the same size and the angles all the same?*Key points:*

- [object Object]

M56

Oral / Practical · Formative Only

What problems might you get if you use a really worn or badly sharpened chain?*Key points:*

- [object Object]

M57

Oral / Practical · Formative Only

How do you dispose of your contaminated chainsaw waste and your litter?*Key points:*

- [object Object]

M58

Oral / Practical · Formative Only

What is the safe working distance between operators when cross cutting?*Key points:*

- [object Object]

M59

Oral / Practical · Formative Only

What bio-security measures might you put in place?*Key points:*

- [object Object]

M60

Oral / Practical · Formative Only

What environmental factors do you need to consider?*Key points:*

- [object Object]

M61

Oral / Practical · Formative Only

In this picture, where is the compression found?*Key points:*

- [object Object]

M62

Oral / Practical · Formative Only

And in this picture, where is the compression?*Key points:*

- [object Object]

M63

Oral / Practical · Formative Only

What are you going to do if you get your saw stuck or pinched?*Key points:*

- [object Object]

M64

Oral / Practical · Formative Only

How are you going to cross cut a piece of timber that's slightly larger than your guidebar?*Key points:*

- [object Object]

M65	Oral / Practical · Formative Only
How are you going to cut timber that's under high tension? <i>Key points:</i> • [object Object]	
M66	Oral / Practical · Formative Only
When would you use a bore cut when cross cutting timber? <i>Key points:</i> • [object Object]	
M67	Oral / Practical · Formative Only
Once you've cut your timber, how are you going to move it safely? <i>Key points:</i> • [object Object]	
M68	Oral / Practical · Formative Only
How do you make moving and working more ergonomic? <i>Key points:</i> • [object Object]	
M69	Oral / Practical · Formative Only
How are you going to make sure your timber stacks are stable? <i>Key points:</i> • [object Object]	
M70	Oral / Practical · Formative Only
What else do you need to think about when stacking lots of different timber? <i>Key points:</i> • [object Object]	
M71	Oral / Practical · Formative Only
Before you start your saw, what checks do you need to do? <i>Key points:</i> • [object Object]	
M72	Oral / Practical · Formative Only
What functional checks do you need to perform before you actually use your saw? <i>Key points:</i> • [object Object]	
M73	Oral / Practical · Formative Only
With all checks done and everything in good condition and working correctly, you are ready to undertake the practical assessment. <i>Key points:</i> • [object Object]	
M74	Oral / Practical · Formative Only
For the cutting part of the assessment you will need to demonstrate multiple cuts under tension and compression and perform bore cuts. <i>Key points:</i> • [object Object]	
M75	Oral / Practical · Formative Only
Good luck. <i>Key points:</i> • [object Object]	

M76

Oral / Practical · Formative Only

What bit of the bar do you need to avoid cutting with?

Key points:

- [object Object]

M77

Oral / Practical · Formative Only

What PPE do you need when using and operating a chainsaw?

Key points:

- [object Object]

M78

Oral / Practical · Formative Only

What do the Manual Handling Operations Regulations 1992 (MHOR) require you to do before manually moving heavy timber on site?

Key points:

- [object Object]

M79

Oral / Practical · Formative Only

How do the Manual Handling Operations Regulations 1992 apply to log handling and timber stacking?

Key points:

- [object Object]

